

7824 Zero Road, second floor

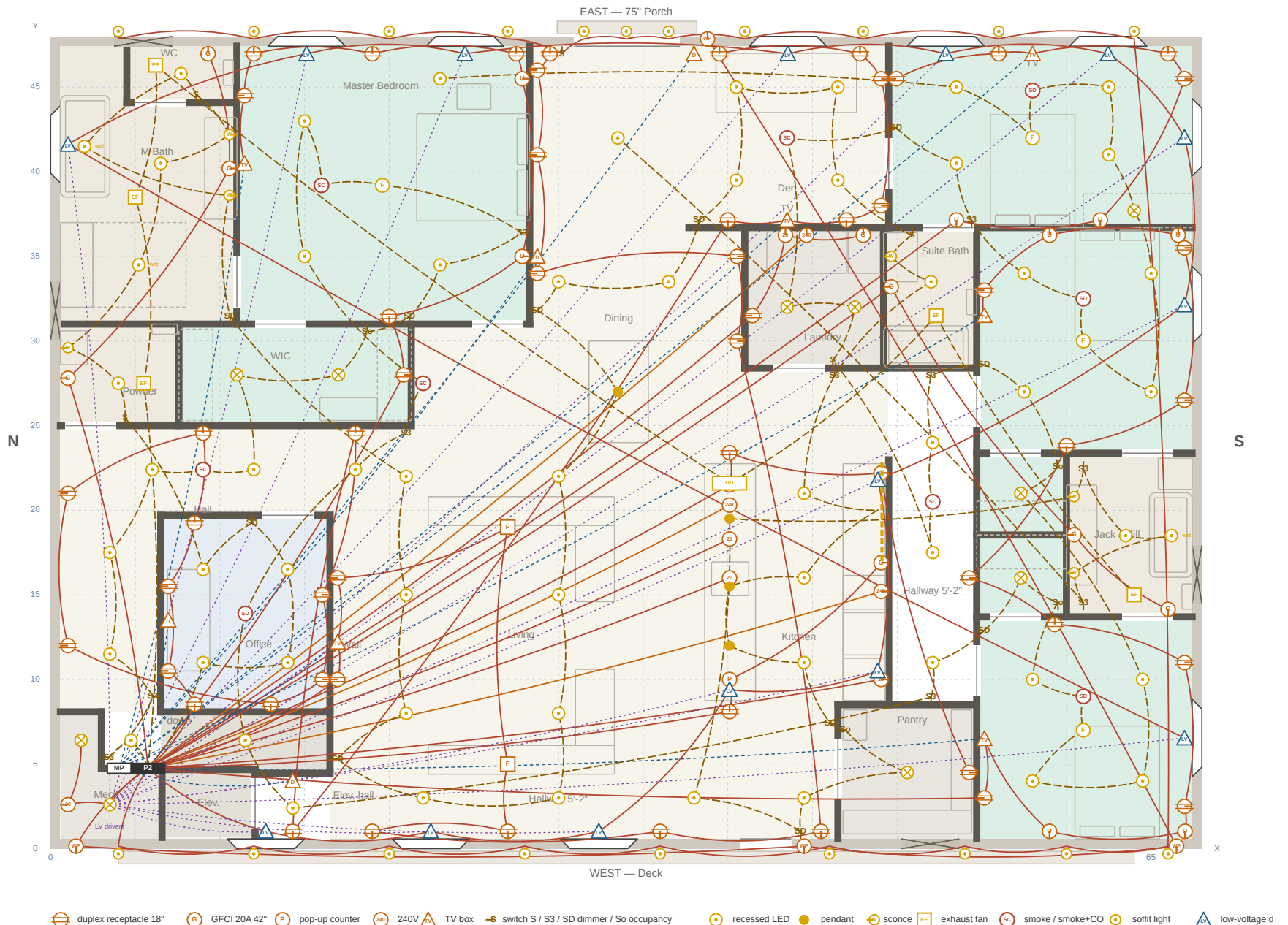
E-1 Electrical plan · packet for the electrical contractor

Devices, luminaires, alarms, low voltage and the circuit schedule for sub-panel P2, with one sheet per circuit.

Rev C · 2026-10-01 · Schematic for pricing and coordination, not a stamped construction document. Verify every dimension on site. Coordinates are feet from the north and west brick faces; the plan is drawn with east at the top and north on the left. Wall numbers match the concept drawing. Questions: name the sheet and the tag (for example E-1 R12).

Contents

1. E-1 sheet
2. Circuit sheets E-1.1 onward and E-1.D low voltage
3. Notes and open items
4. Schedules
5. Bill of materials



7824 Zero Road — second floor

Casper, Wyoming · 68' x 48' to brick, faces 7" inside · 9' ceiling · grid 5'
Wall numbers as the floor plan drawing of 2026-09-29

Electrical plan

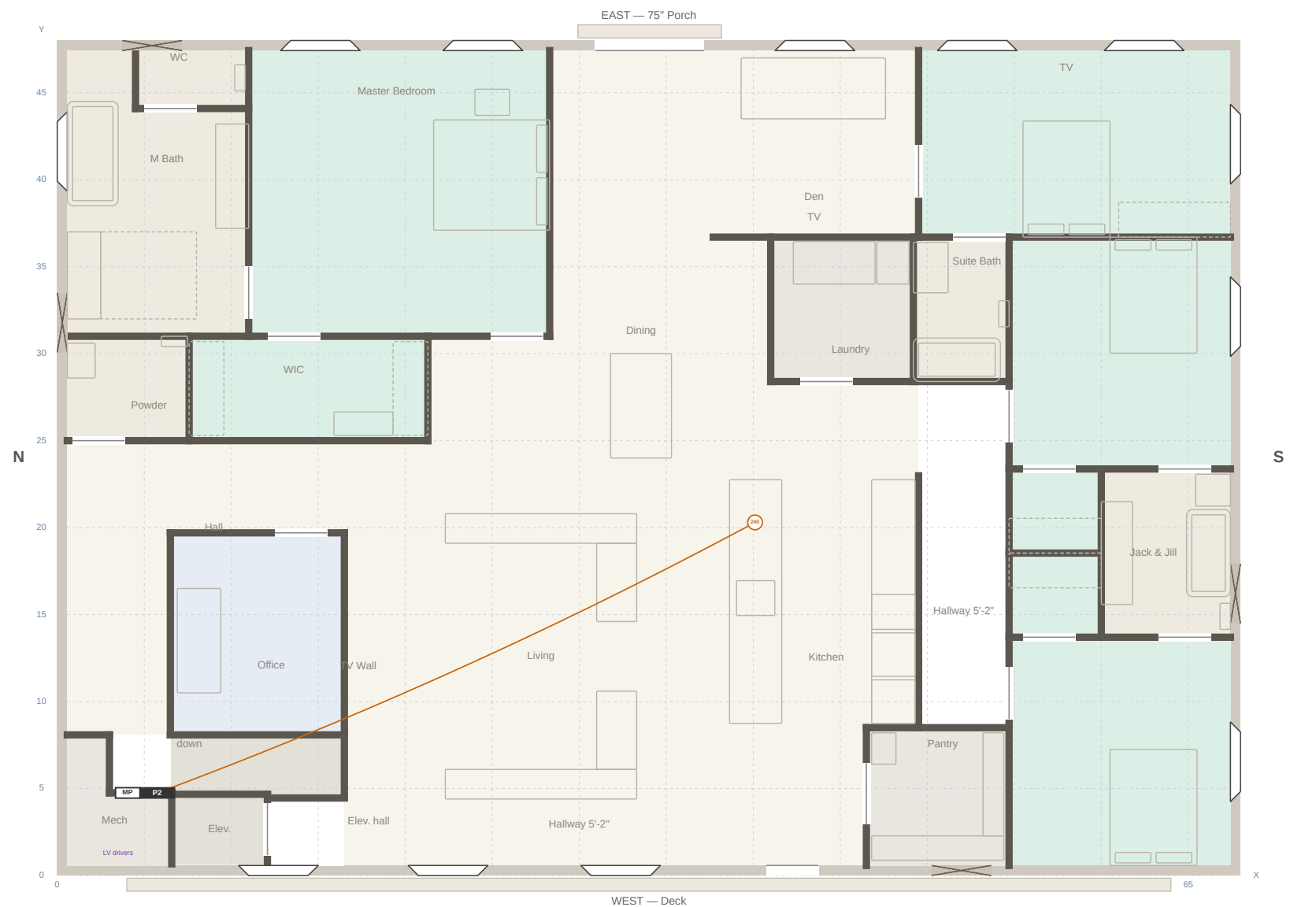
Devices, luminaires, alarms and the circuit schedule for sub-panel P2

Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson

E-1

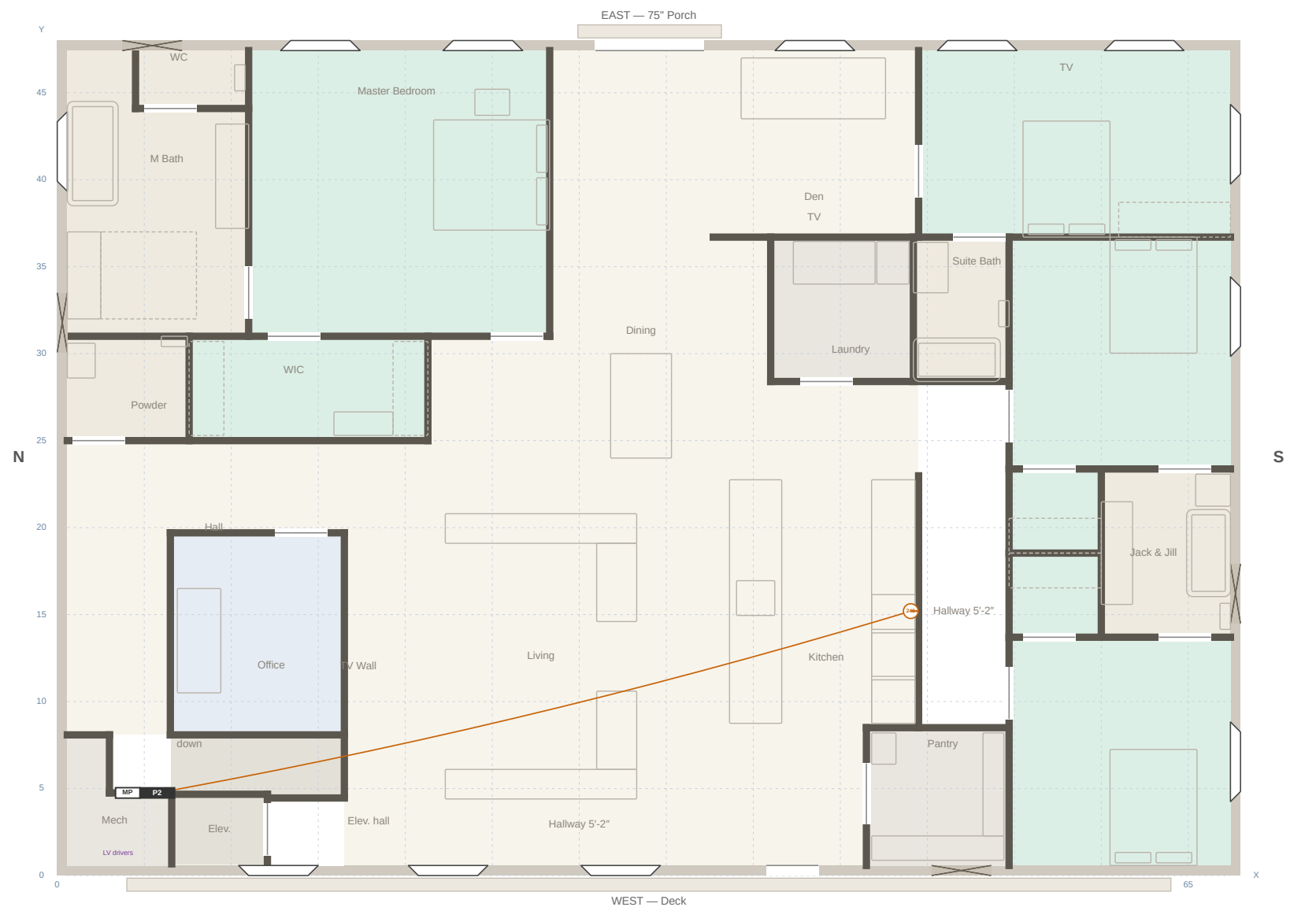
rev C · 2026-10-01

E-1.1 — circuit 1-2: Induction cooktop, 240V 40A 2P, #8 Cu · kitchen wall run



duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.2 — circuit 3-4: Wall oven, side-opening, 240V 30A 2P, #10 Cu · kitchen wall run

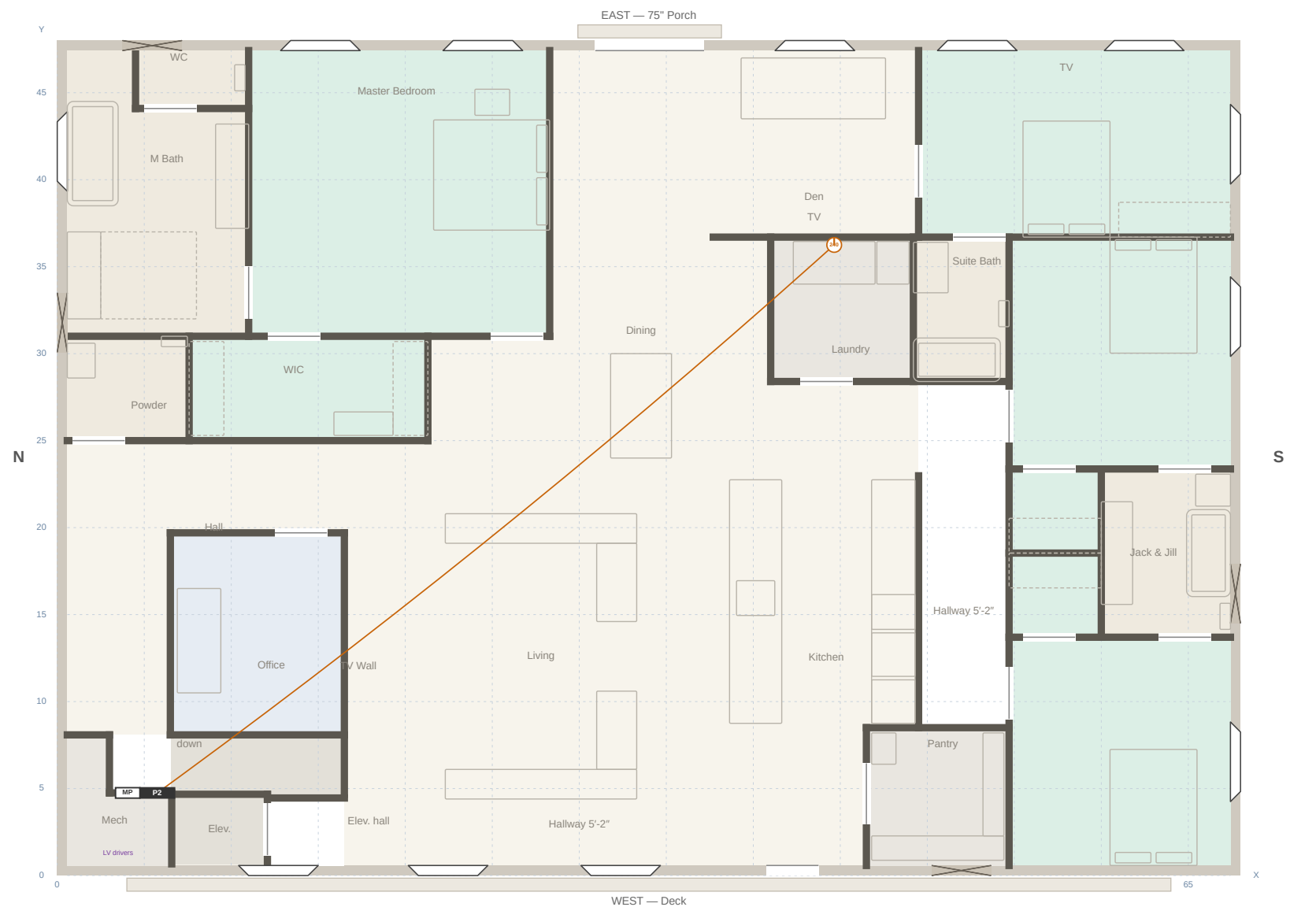


duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor		Electrical plan	
Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5'		Devices, luminaires, alarms and the circuit schedule for sub-panel P2	
Wall numbers as the floor plan drawing of 2026-09-29		Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	

E-1

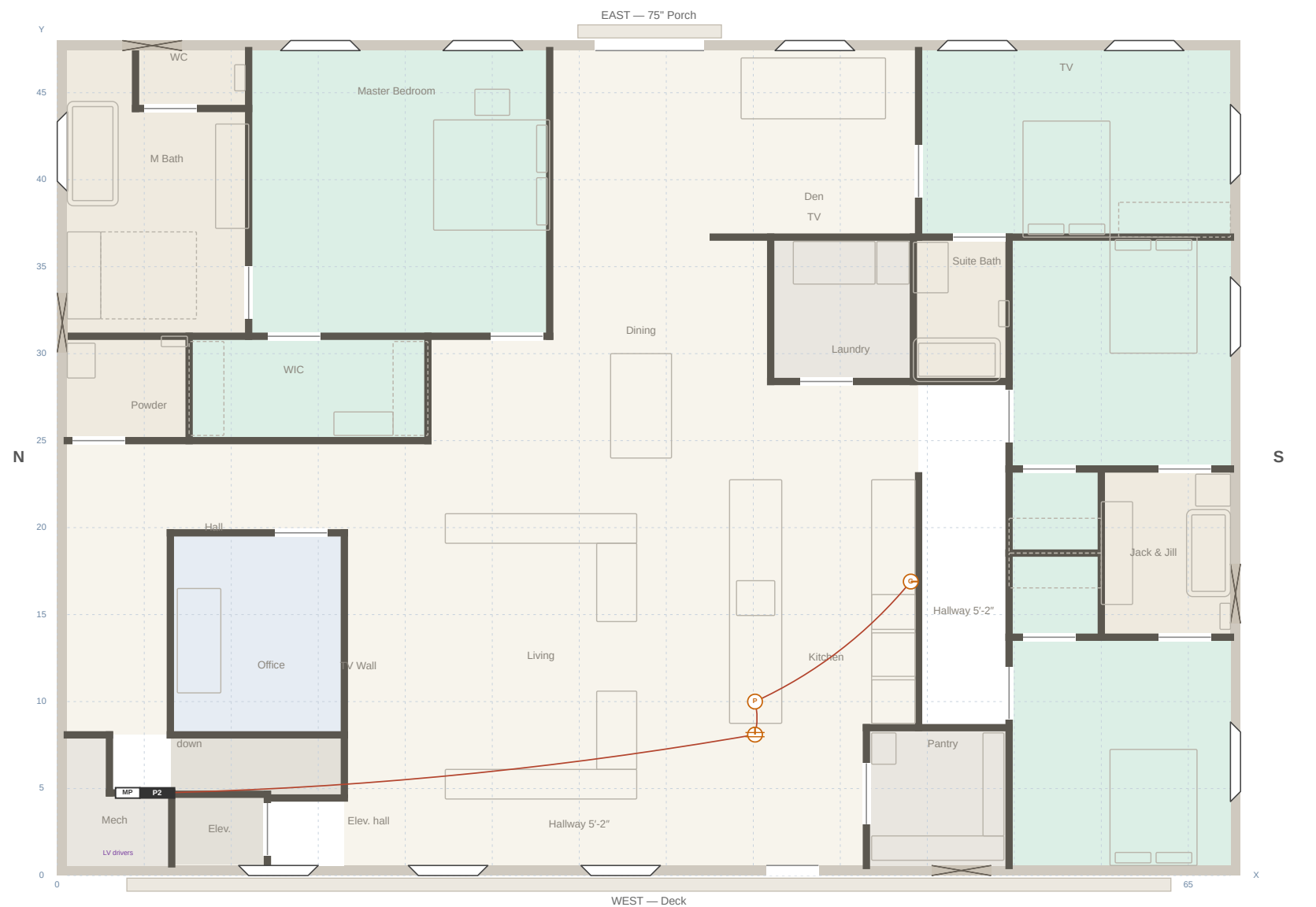
rev C · 2026-10-01

E-1.3 — circuit 5-6: Dryer, 240V 30A 2P, #10 Cu · laundry



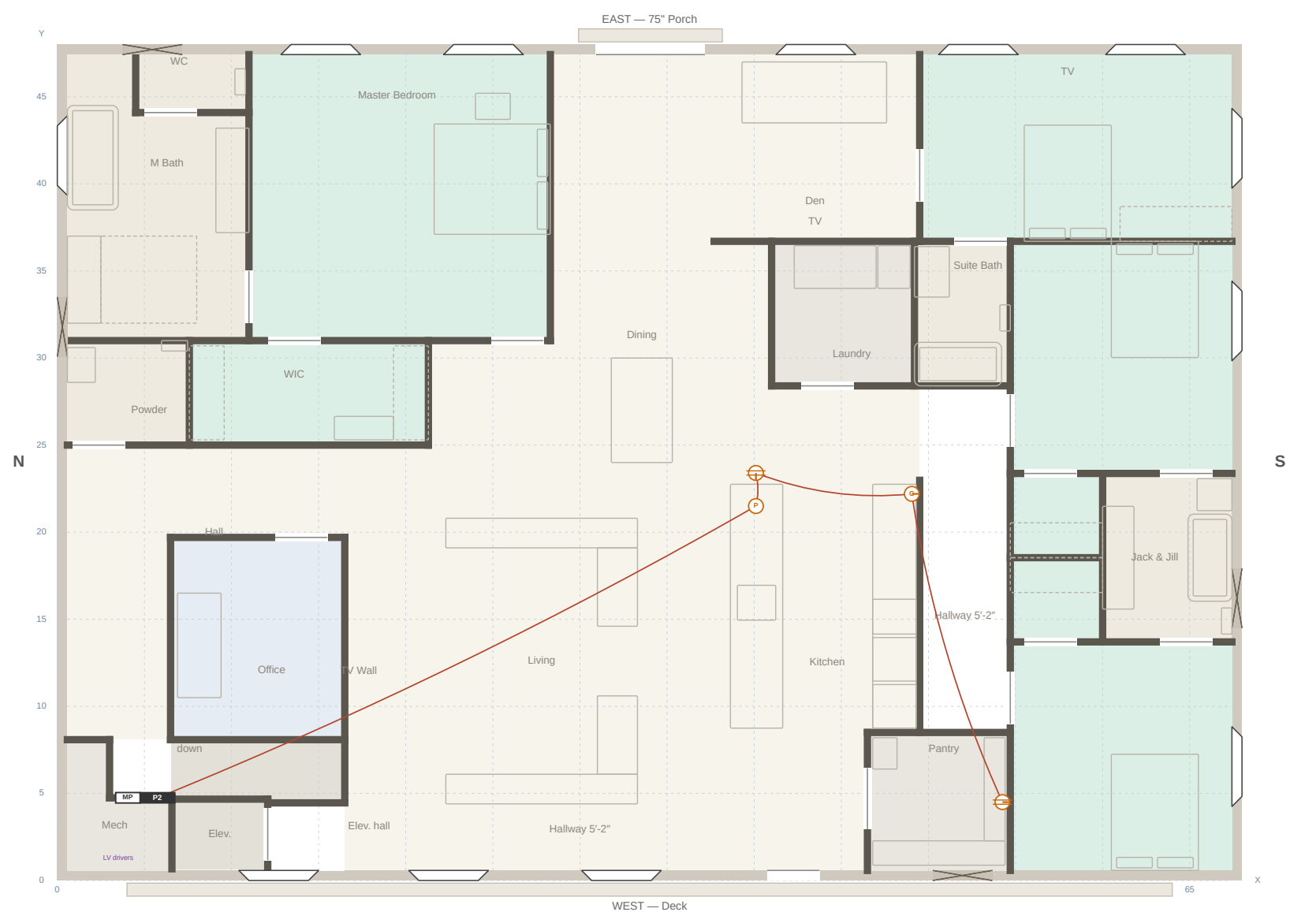
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.4 — circuit 7: Kitchen small appliance 1, 20A GFCI/AFCI, #12 · island sink end, wall run



duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant scone exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.5 — circuit 8: Kitchen small appliance 2 + pantry, 20A GFCI/AFCI, #12 · island prep end, wall run, pantry

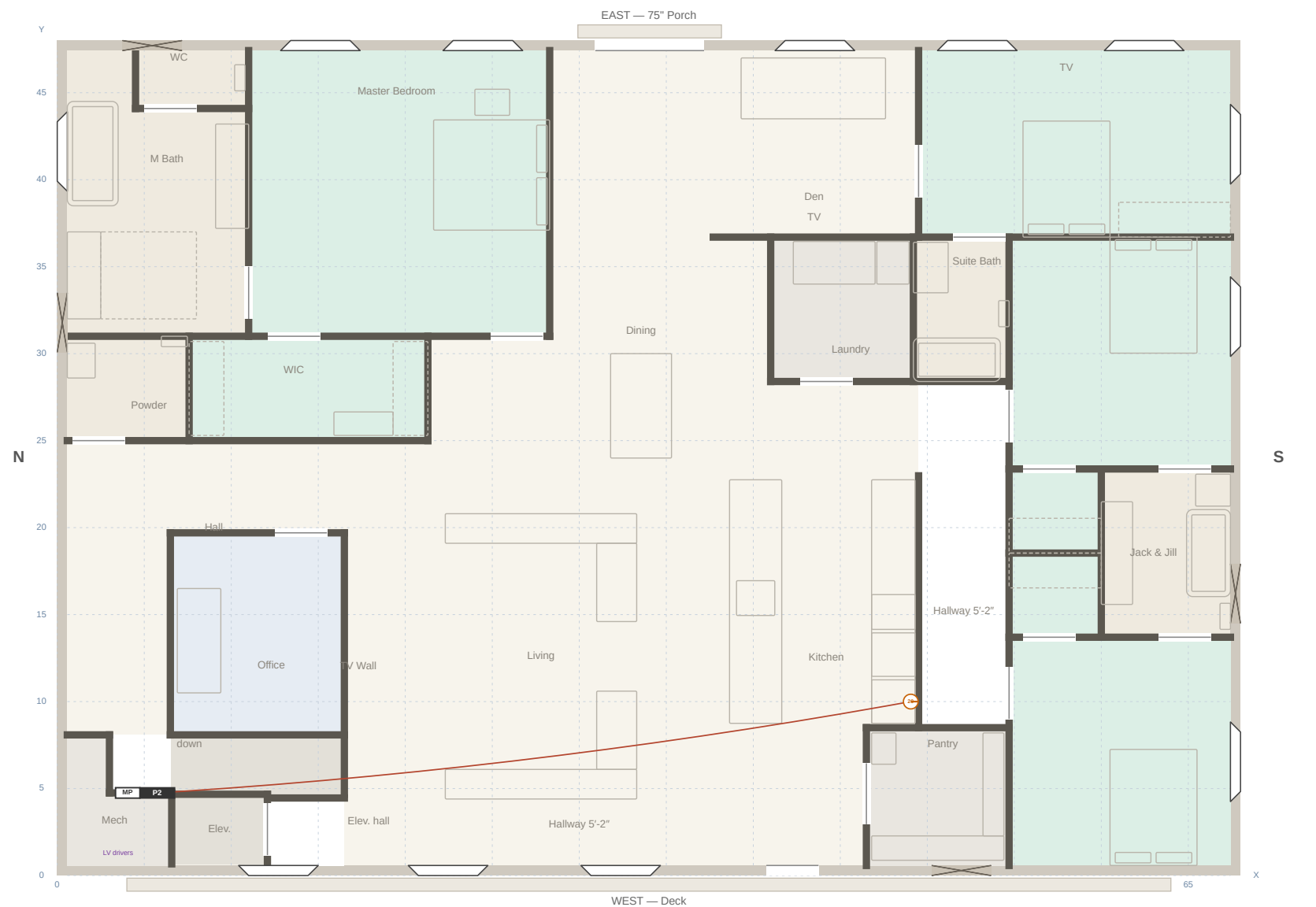


duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor		Electrical plan	
Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5'		Devices, luminaires, alarms and the circuit schedule for sub-panel P2	
Wall numbers as the floor plan drawing of 2026-09-29		Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	

E-1

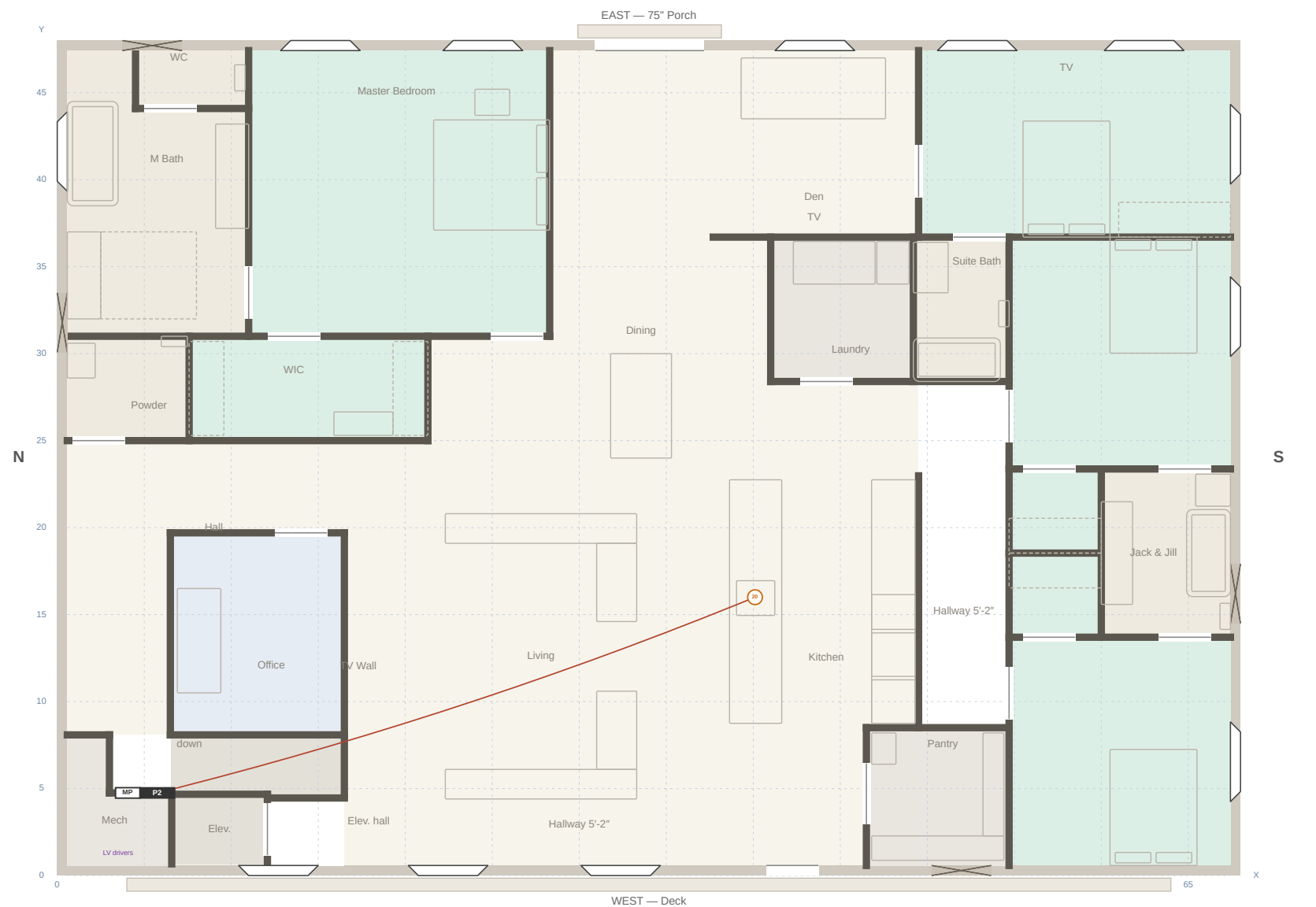
rev C · 2026-10-01

E-1.6 — circuit 9: Refrigerator, 20A AFCI, #12 · kitchen



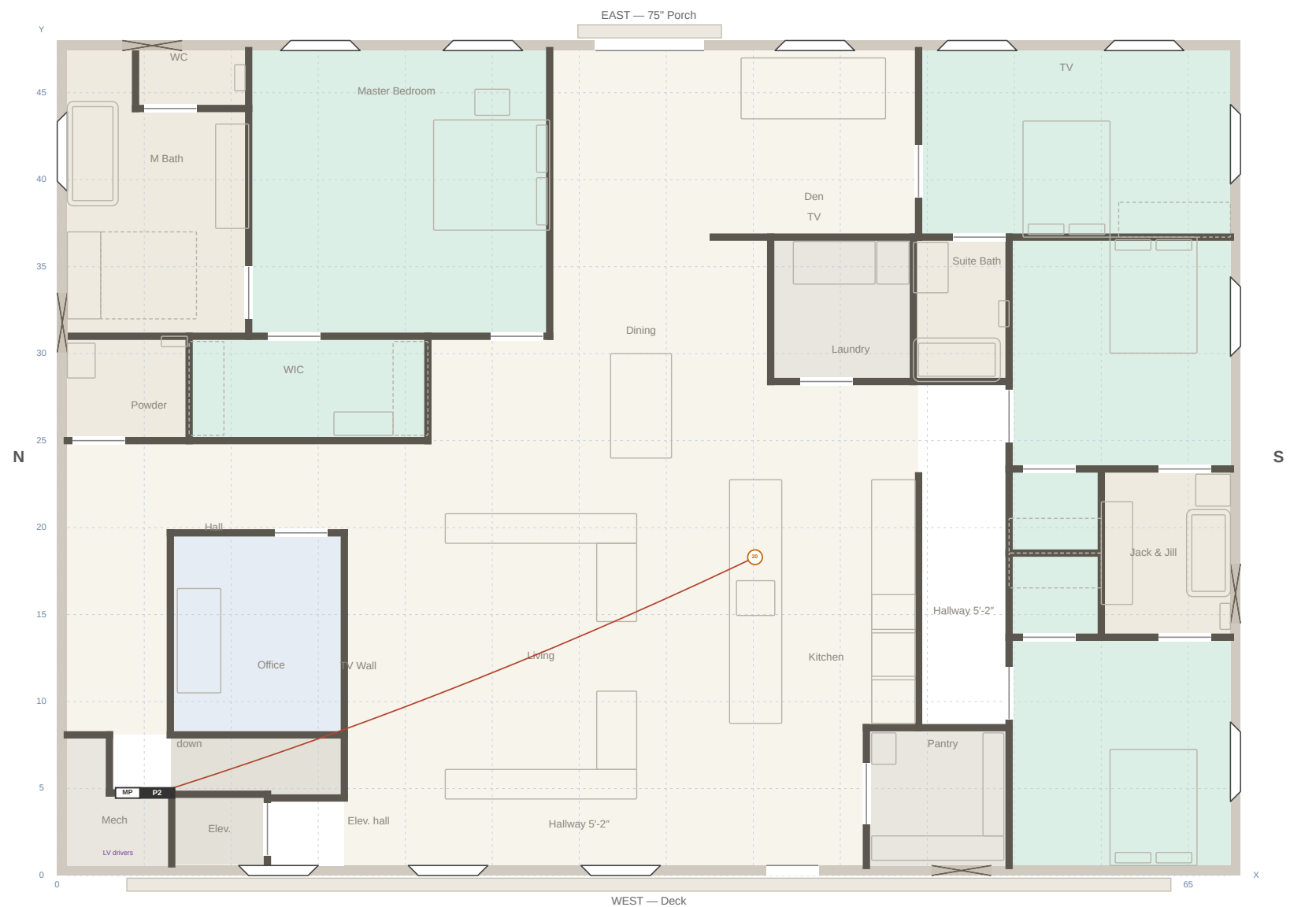
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.8 — circuit 11: Dishwasher drawers, 20A GFCI/AFCI, #12 · island



duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant scone exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.9 — circuit 12: Microwave drawer, 20A AFCI, #12 · island



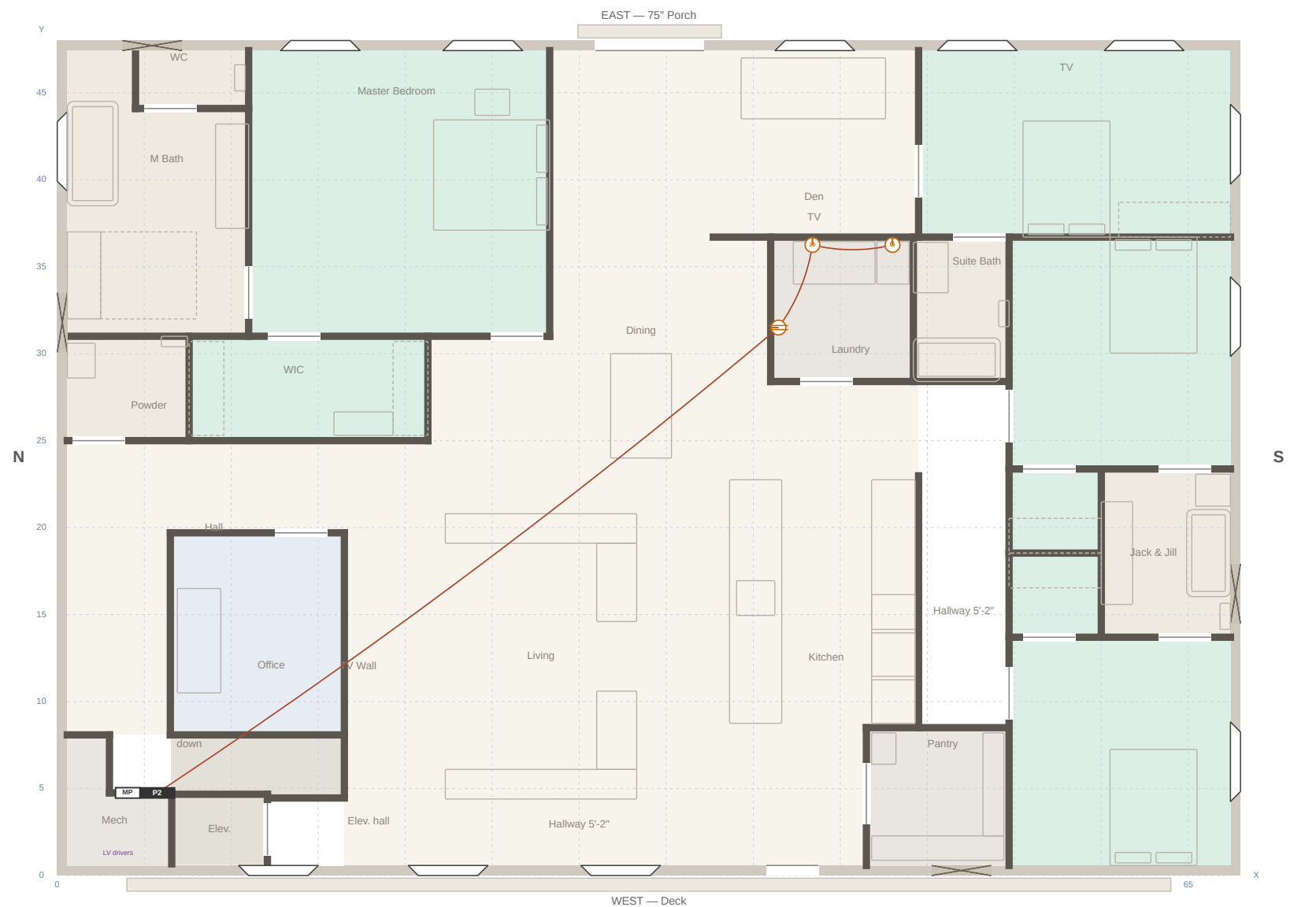
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d	
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29	Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson

E-1

rev C · 2026-10-01

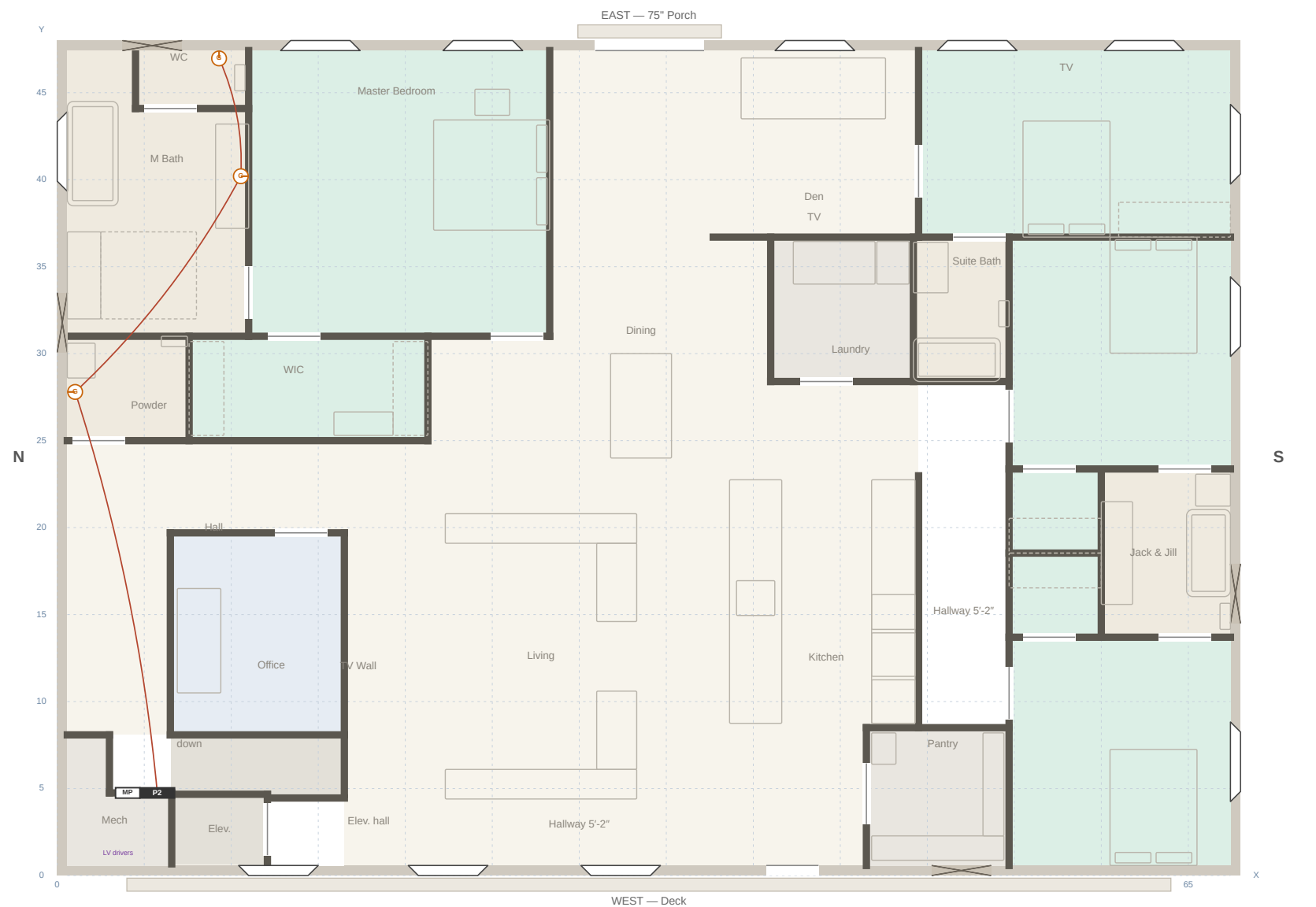
drawn for K. Christianson

E-1.10 — circuit 13: Laundry: washer, sink, counter, 20A GFCI/AFCI, #12 · laundry



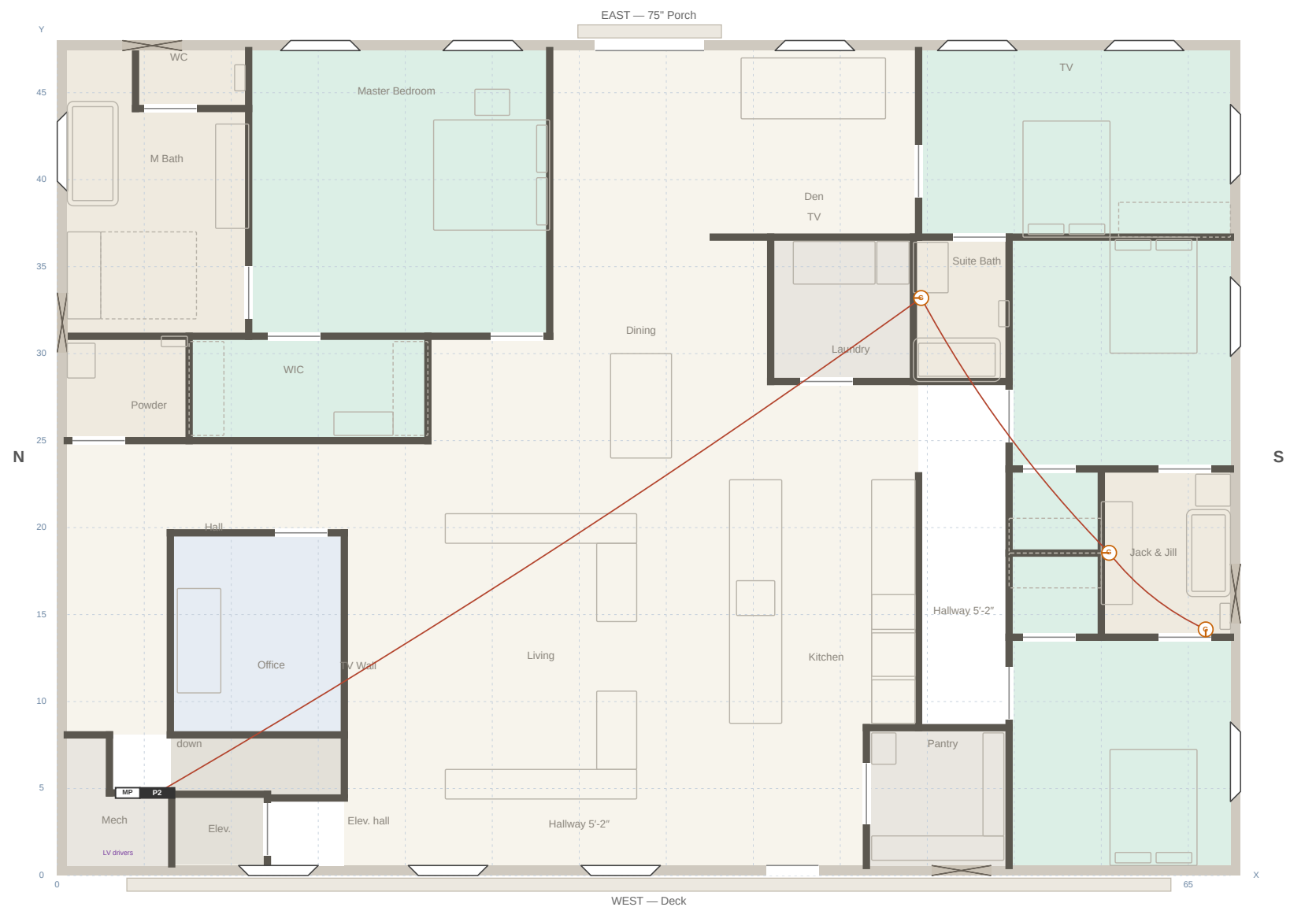
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.11 — circuit 14: Bath receptacles west, 20A GFCI, #12 · master bath, WC, powder



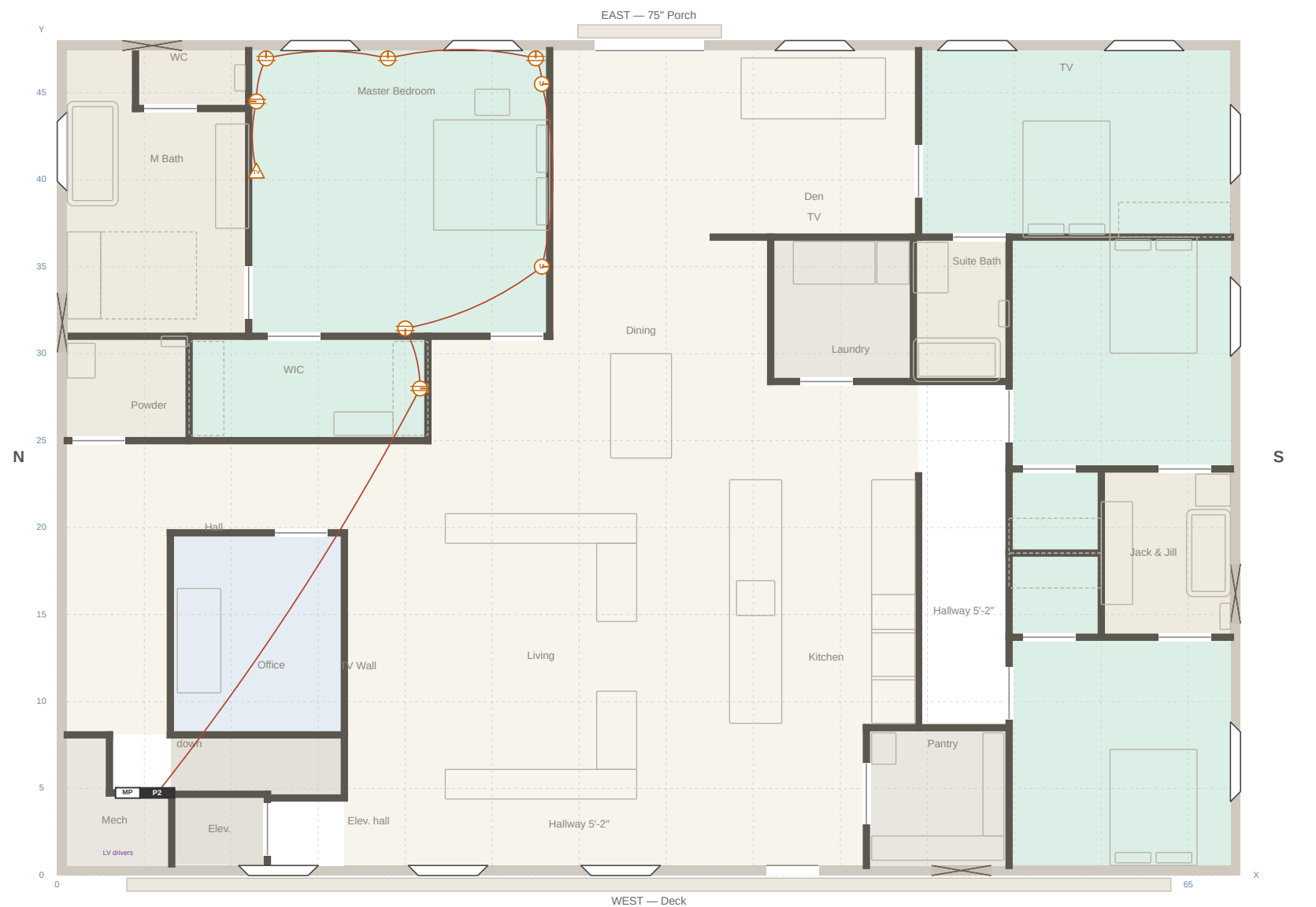
duplex receptacle 18"		GFCI 20A 42"	pop-up counter	240V	TV box	switch S / S3 / SD dimmer / So occupancy	recessed LED	pendant	sconce	exhaust fan	smoke / smoke+CO	soffit light	low-voltage d
7824 Zero Road — second floor						Electrical plan				E-1			
Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5'						Devices, luminaires, alarms and the circuit schedule for sub-panel P2				rev C · 2026-10-01			
Wall numbers as the floor plan drawing of 2026-09-29						Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson				drawn for K. Christianson			

E-1.12 — circuit 15: Bath receptacles east, 20A GFCI, #12 · suite bath, Jack & Jill



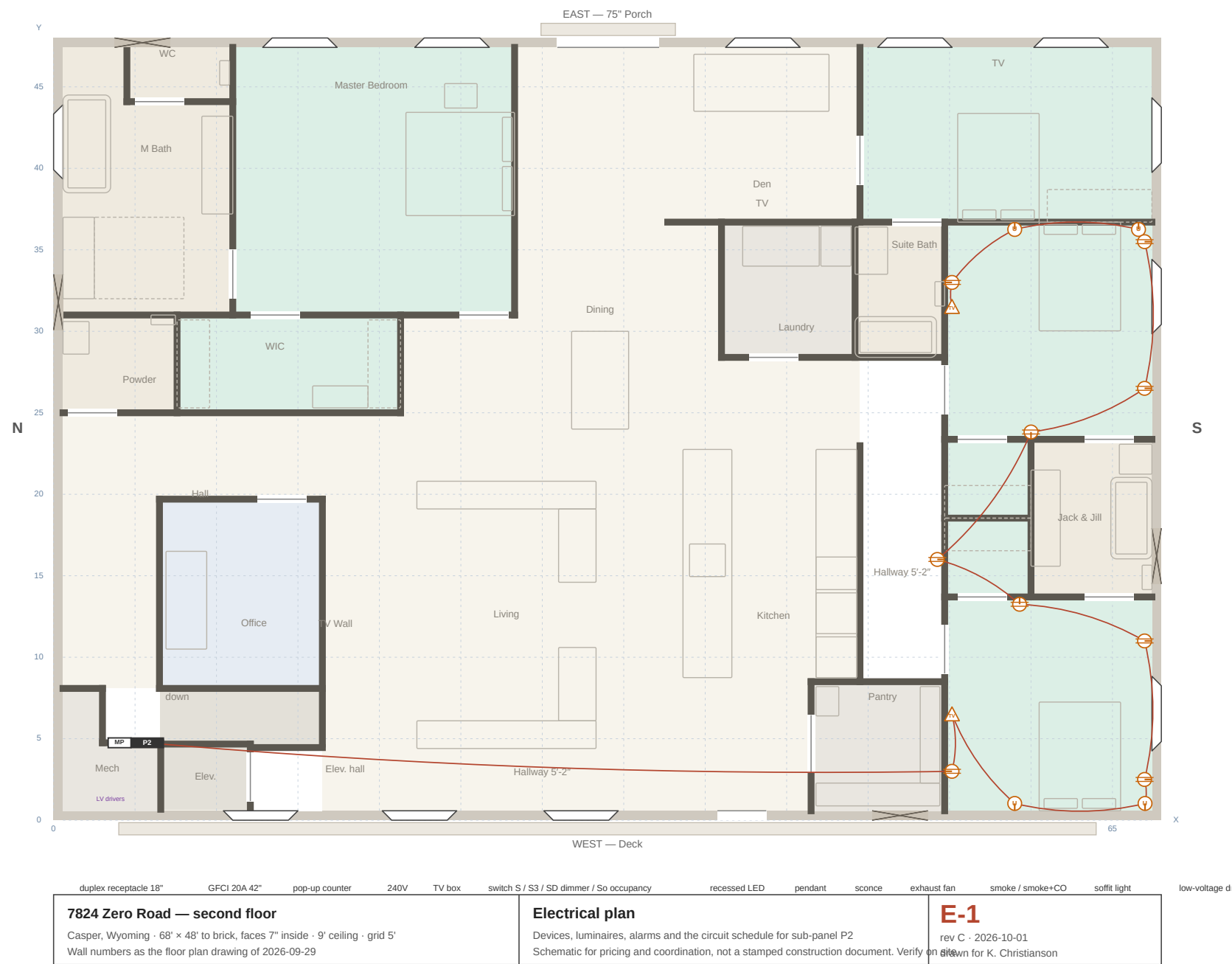
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.13 — circuit 16: Master suite receptacles, 15A AFCI, #14 · master bedroom, walk-in

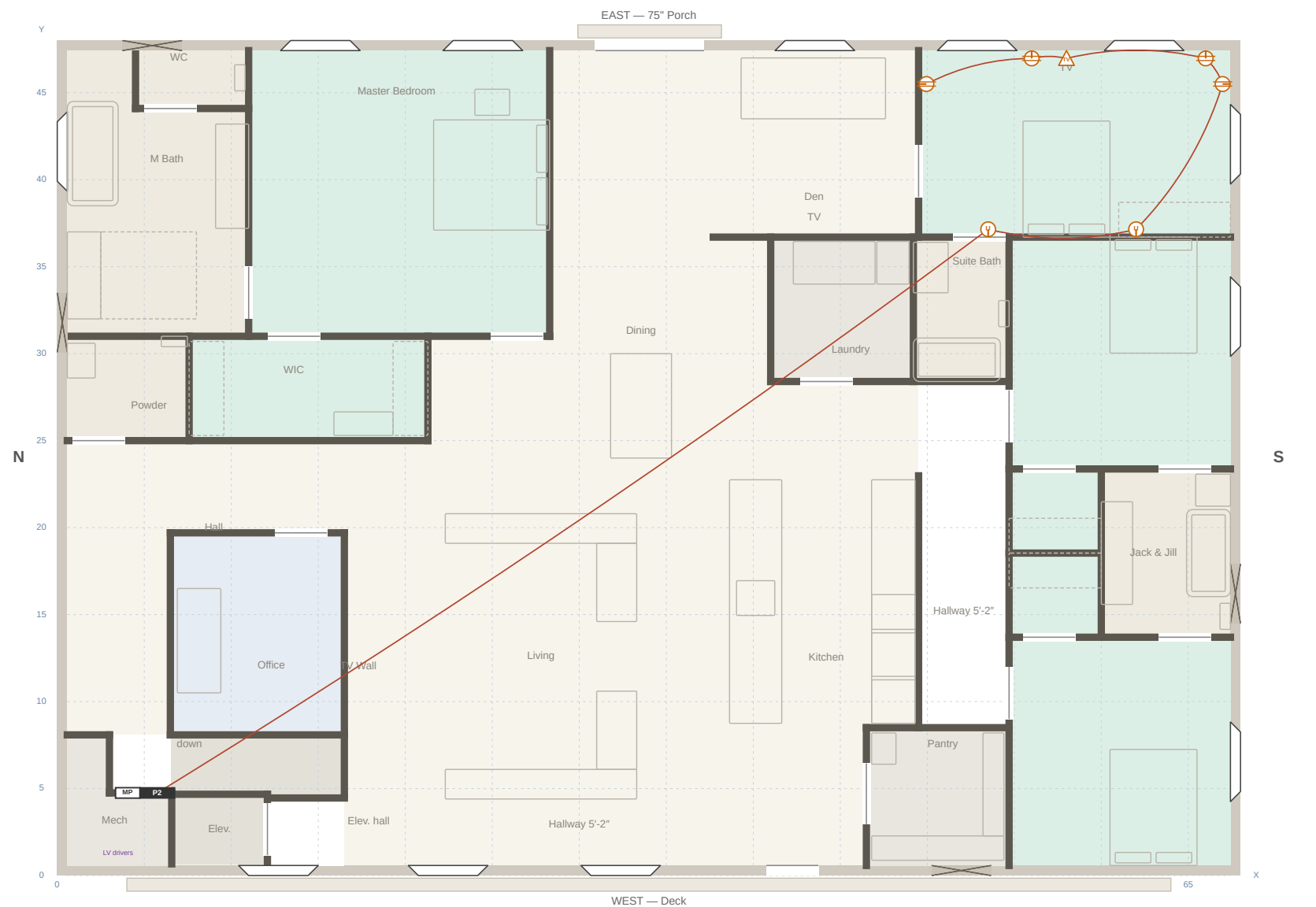


duplex receptacle 18"	GFCI 20A 42"	pop-up counter	240V	TV box	switch S / S3 / SD dimmer / So occupancy	recessed LED	pendant	sconce	exhaust fan	smoke / smoke+CO	soffit light	low-voltage d
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29						Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson			E-1 rev C · 2026-10-01 drawn for K. Christianson			

E-1.14 — circuit 17: Bedrooms 3 and 4 receptacles, 15A AFCI, #14 · bedrooms 3, 4, closets, wing hall

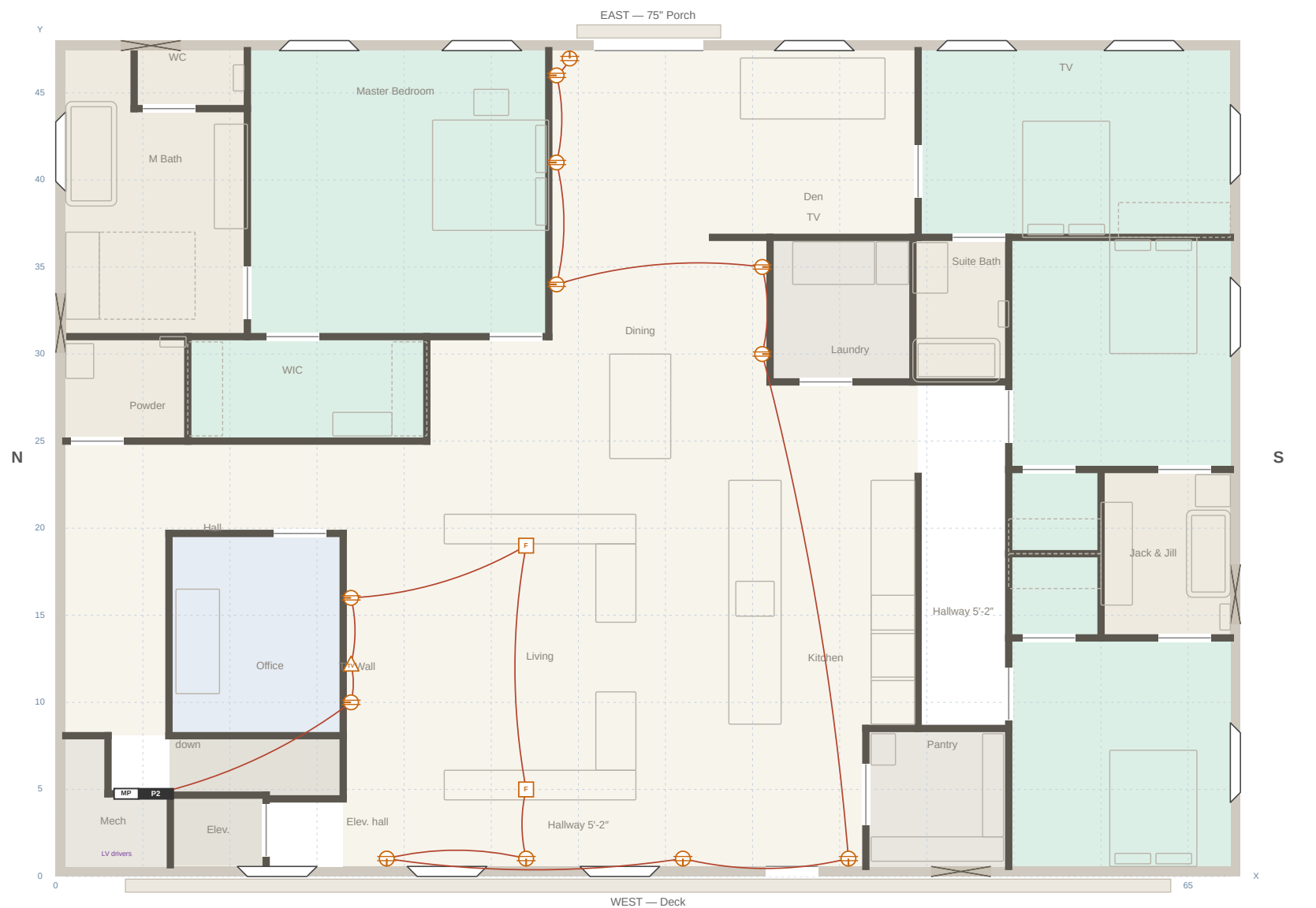


E-1.15 — circuit 18: 2nd Suite receptacles, 15A AFCI, #14 · 2nd Suite



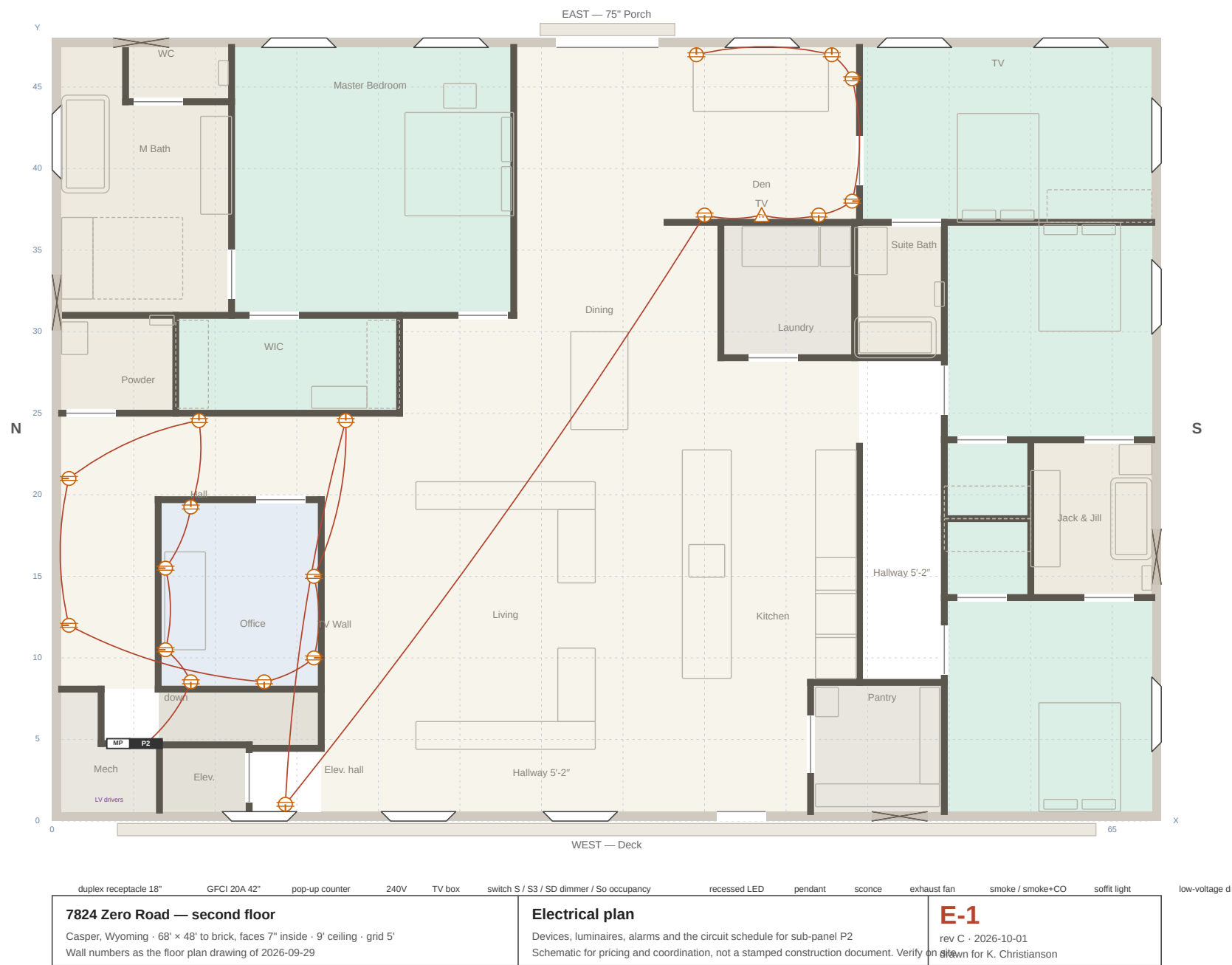
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.16 — circuit 19: Living and dining receptacles, 15A AFCI, #14 · living, hallway, dining, east entry

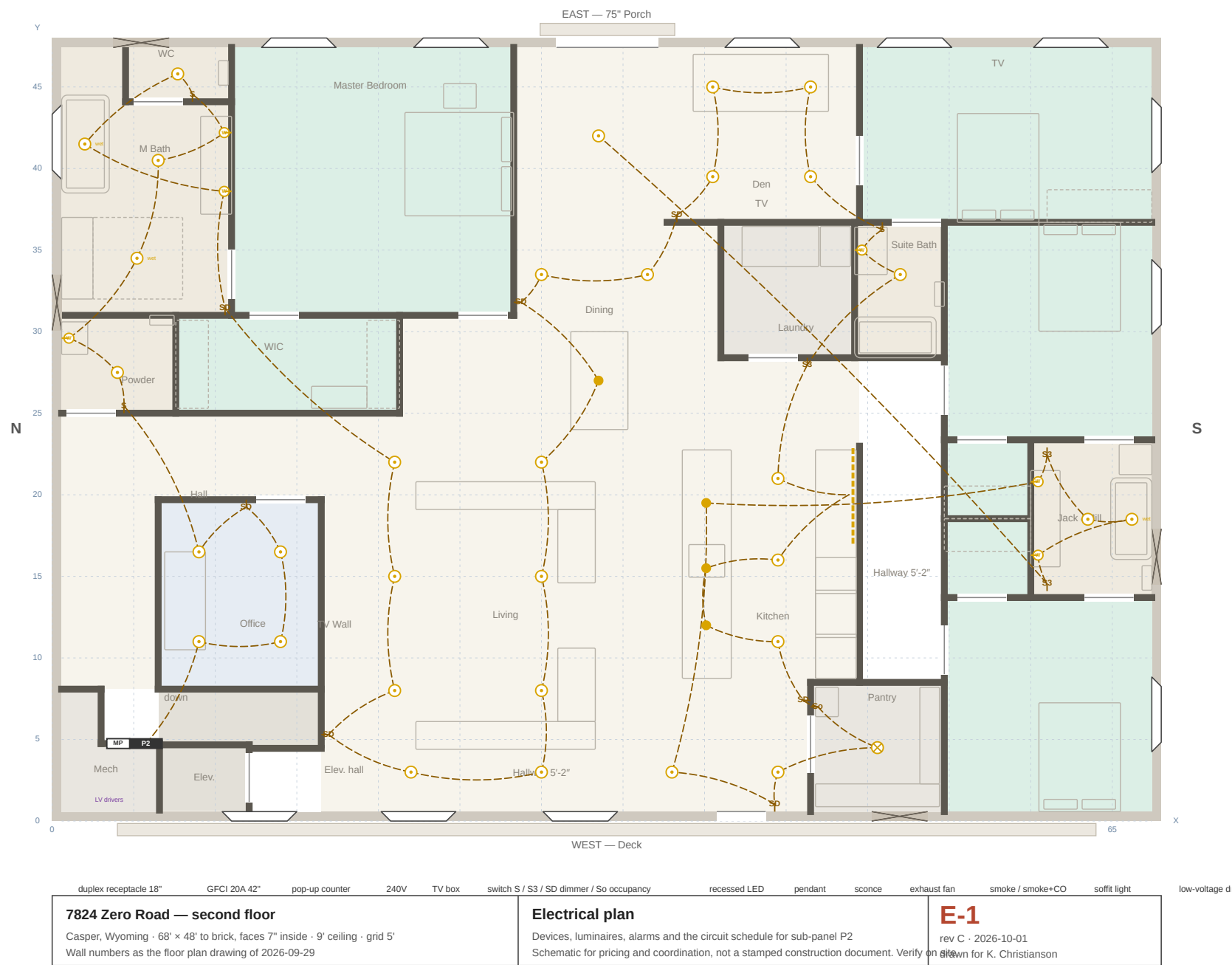


duplex receptacle 18"	GFCI 20A 42"	pop-up counter	240V	TV box	switch S / S3 / SD dimmer / So occupancy	recessed LED	pendant	sconce	exhaust fan	smoke / smoke+CO	soffit light	low-voltage d
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29						Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson			E-1 rev C · 2026-10-01 drawn for K. Christianson			

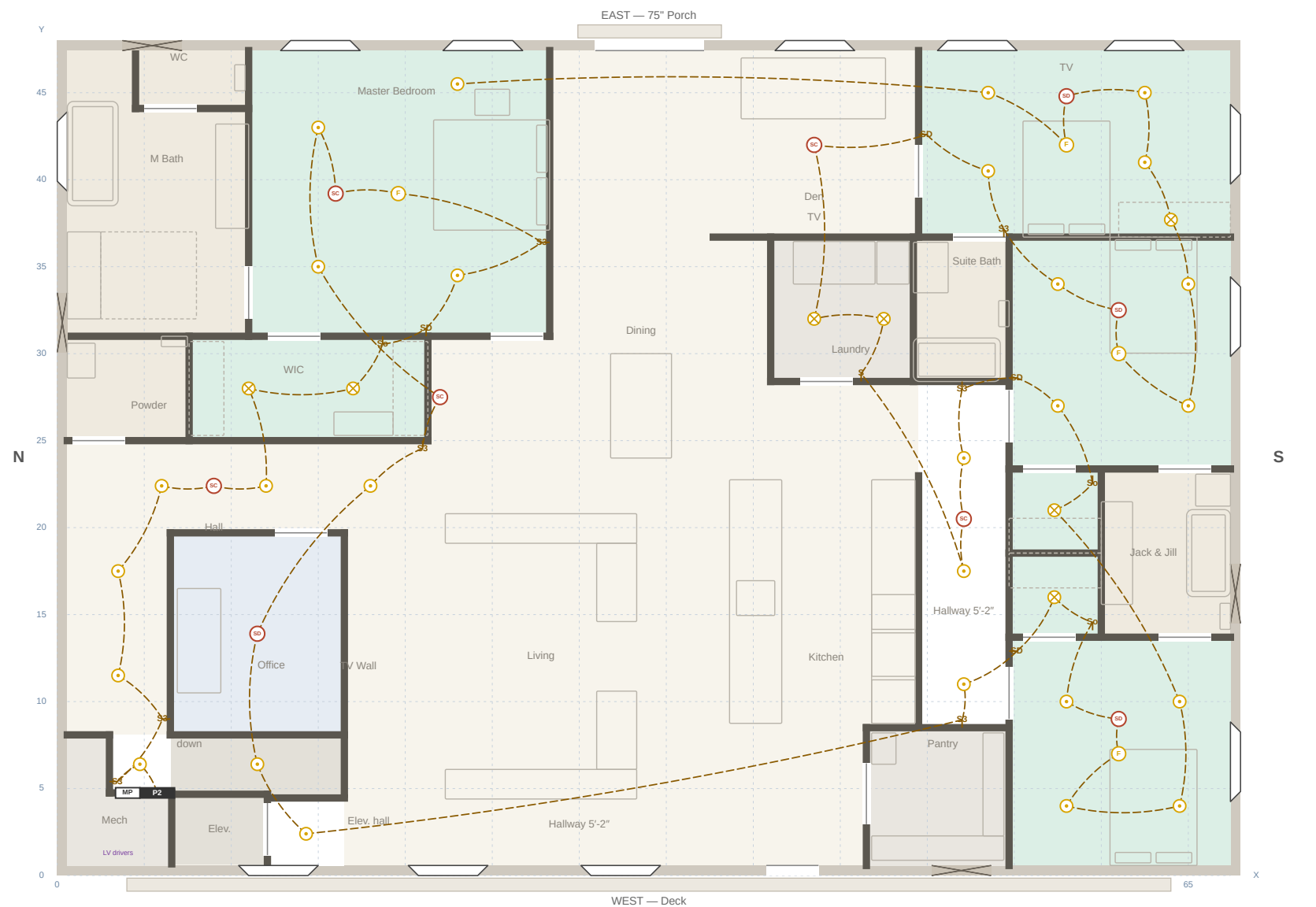
E-1.17 — circuit 20: Den, office, passage receptacles, 15A AFCI, #14 · den, office, passage, elevator hall



E-1.18 — circuit 21: Lighting A, 15A AFCI, #14 · living, dining, kitchen, den, office, baths

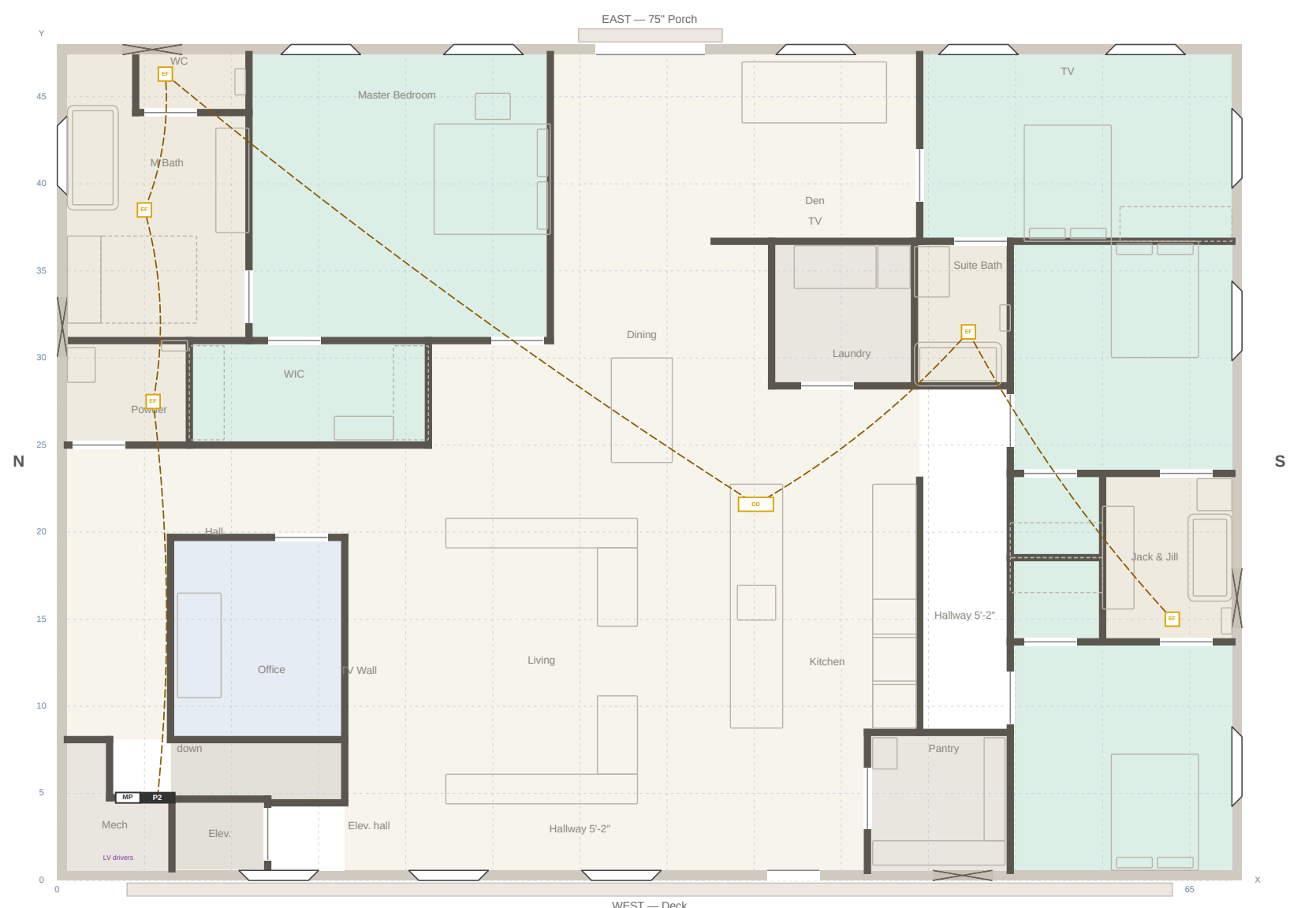


E-1.19 — circuit 22: Lighting B + alarms, 15A AFCI, #14 · bedrooms, closets, halls, stair, smoke/CO alarms



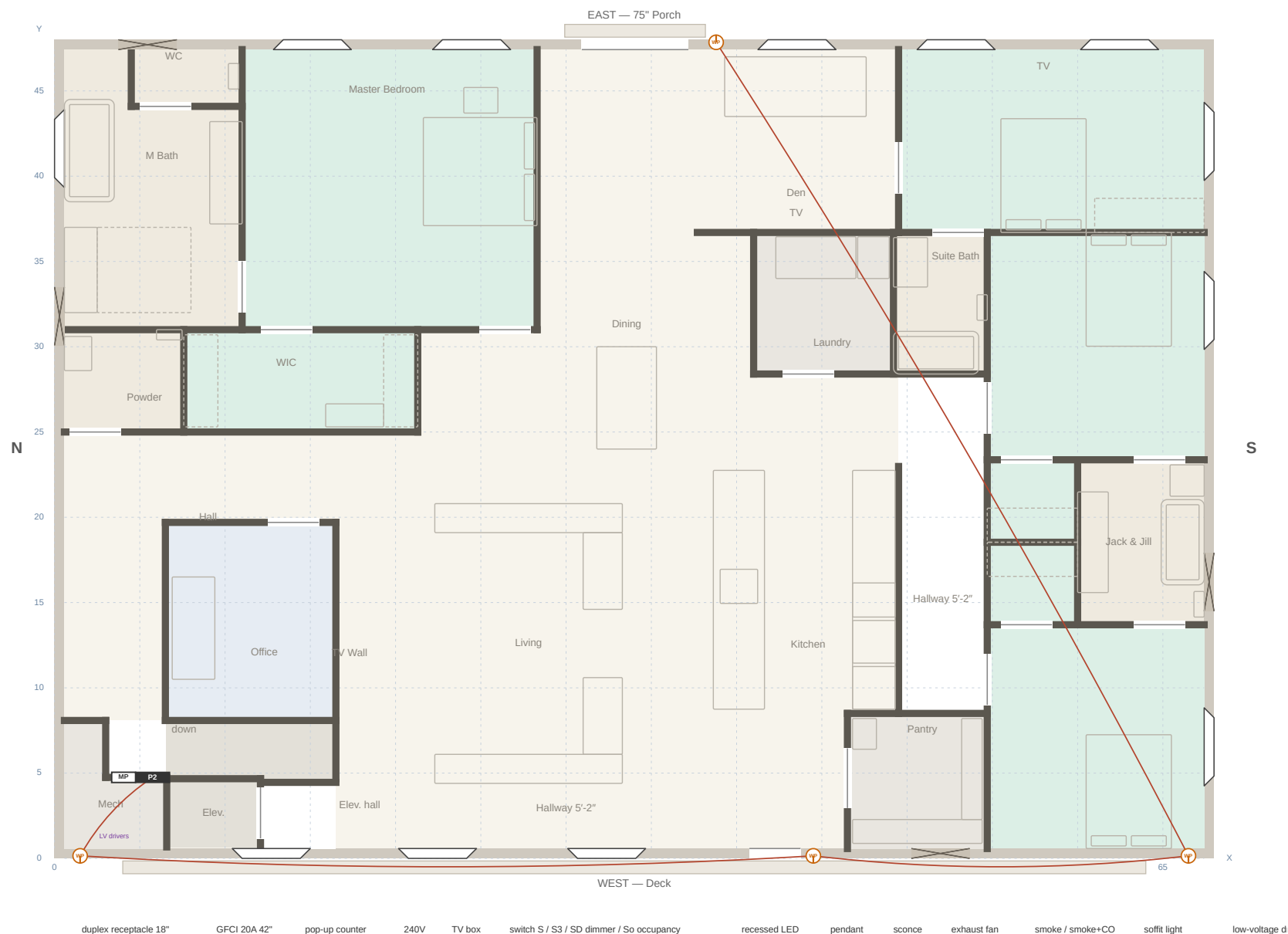
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d 7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29	Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	E-1 rev C · 2026-10-01 drawn by K. Christianson
---	---	---

E-1.20 — circuit 23: Exhaust fans + hood, 15A AFCI, #14 · five bath fans, range hood



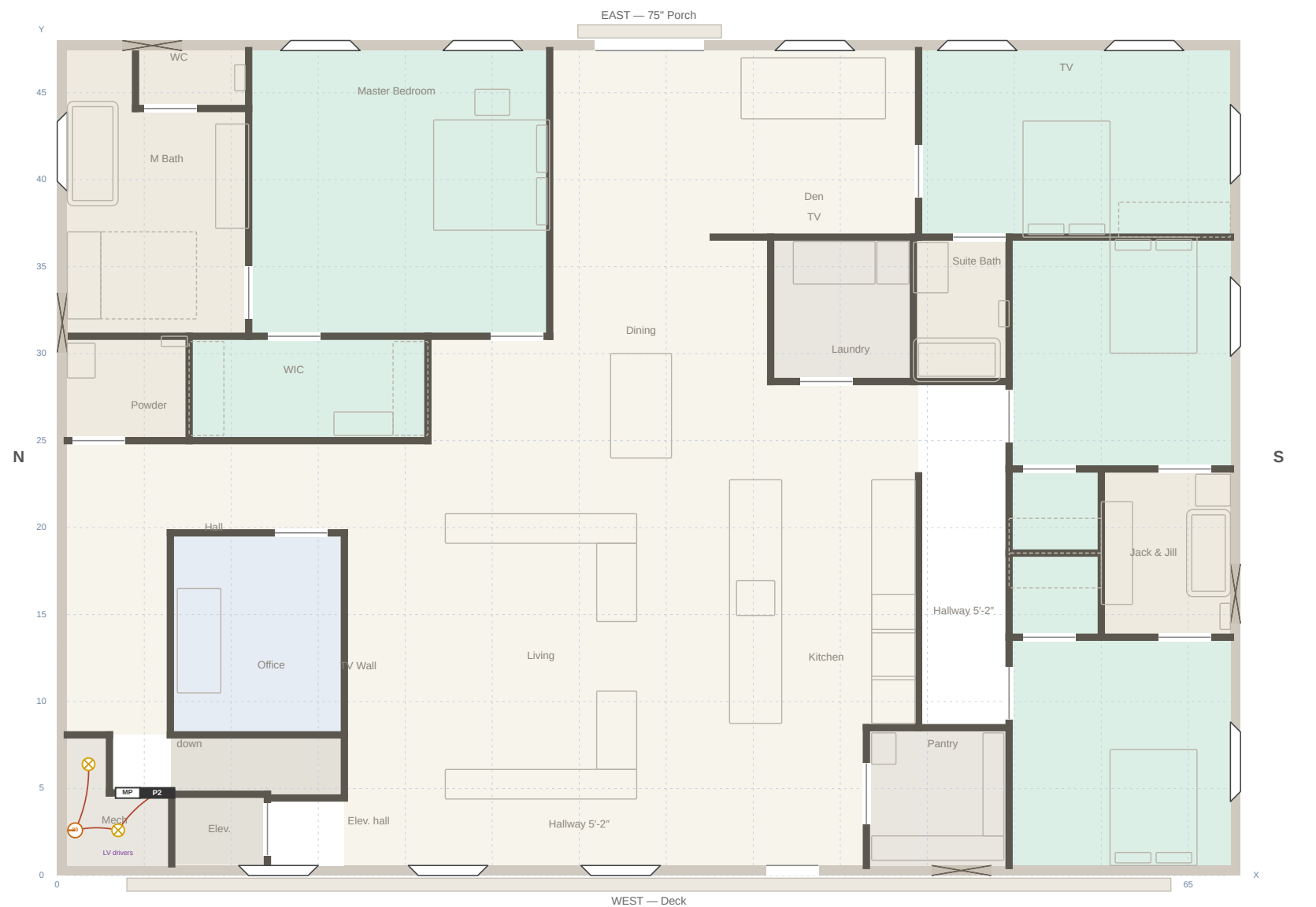
duplex receptacle 18"	GFCI 20A 42"	pop-up counter	240V	TV box	switch S / S3 / SD dimmer / So occupancy	recessed LED	pendant	sconce	exhaust fan	smoke / smoke+CO	soffit light	low-voltage d
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29						Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson			E-1 rev C · 2026-10-01 drawn for K. Christianson			

E-1.21 — circuit 24: Exterior receptacles, 20A GFCI, #12 · soffit receptacles, west and east

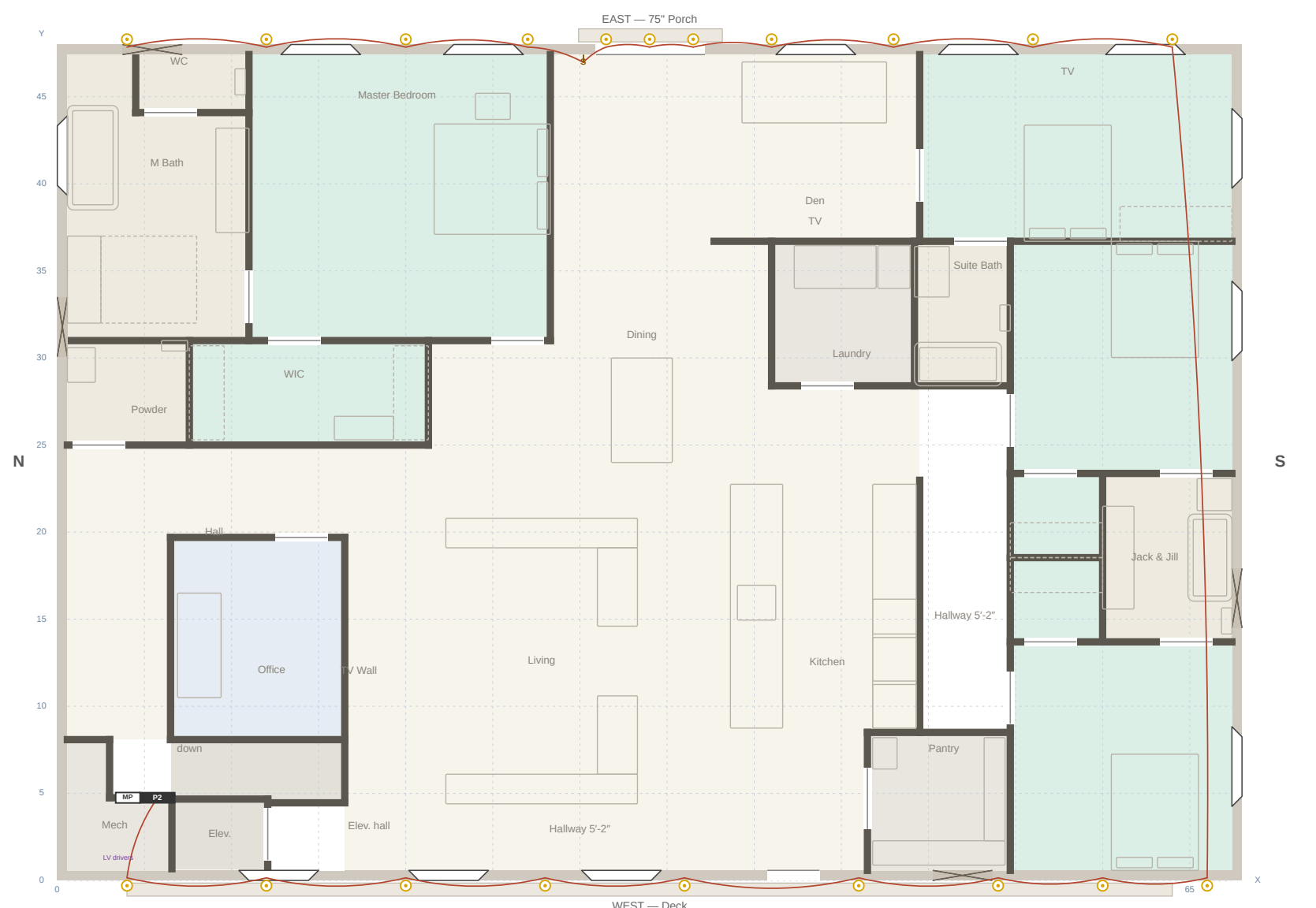


7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29	Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	E-1 rev C · 2026-10-01 drawn for K. Christianson
--	--	---

E-1.22 — circuit 25: Mech closet: ventilation unit, 15A, #14 · ERV / vents, service receptacle

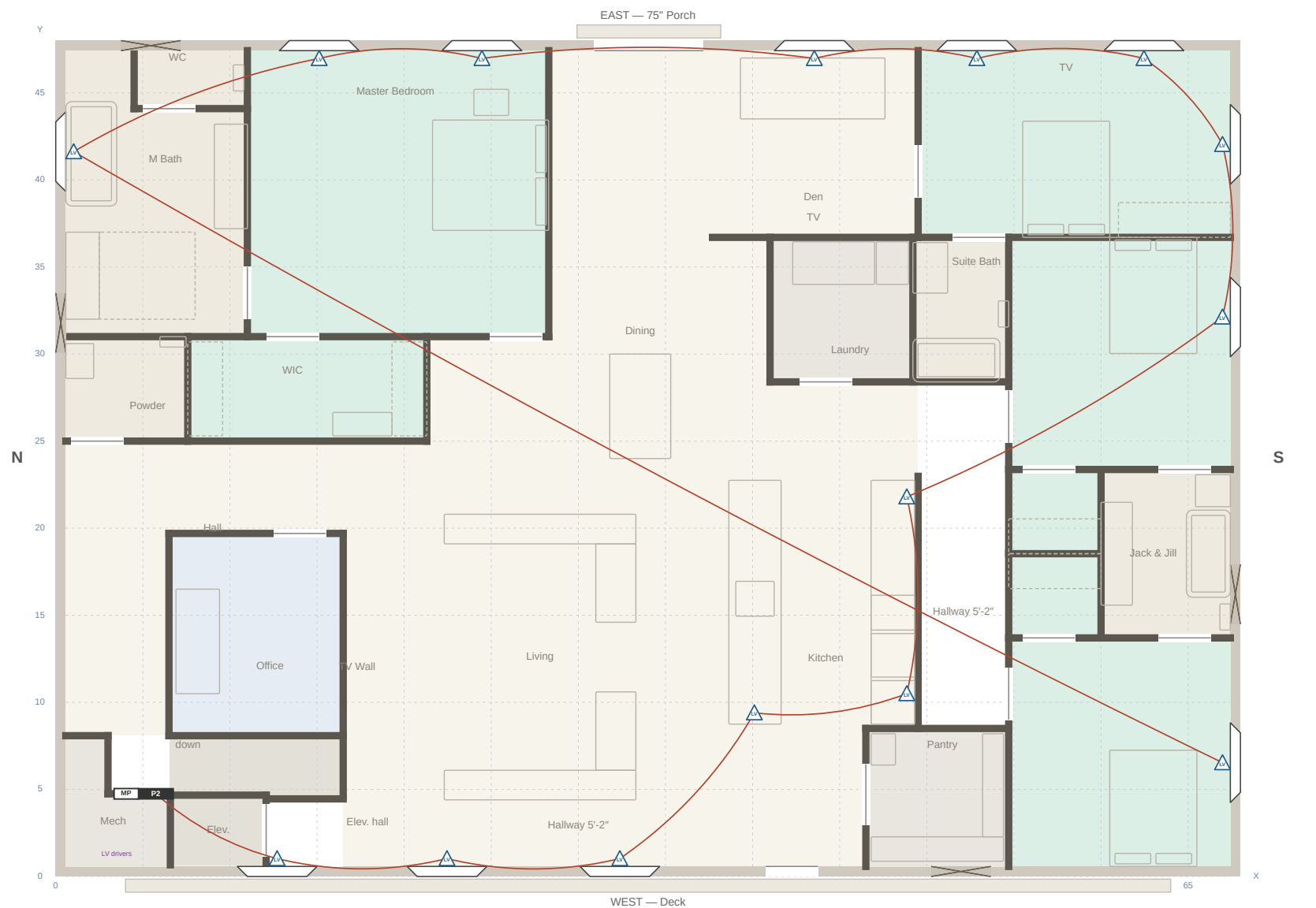


E-1.23 — circuit 26: Exterior soffit lighting, 15A, #14 · west soffit, east door group, east soffit; three switch legs



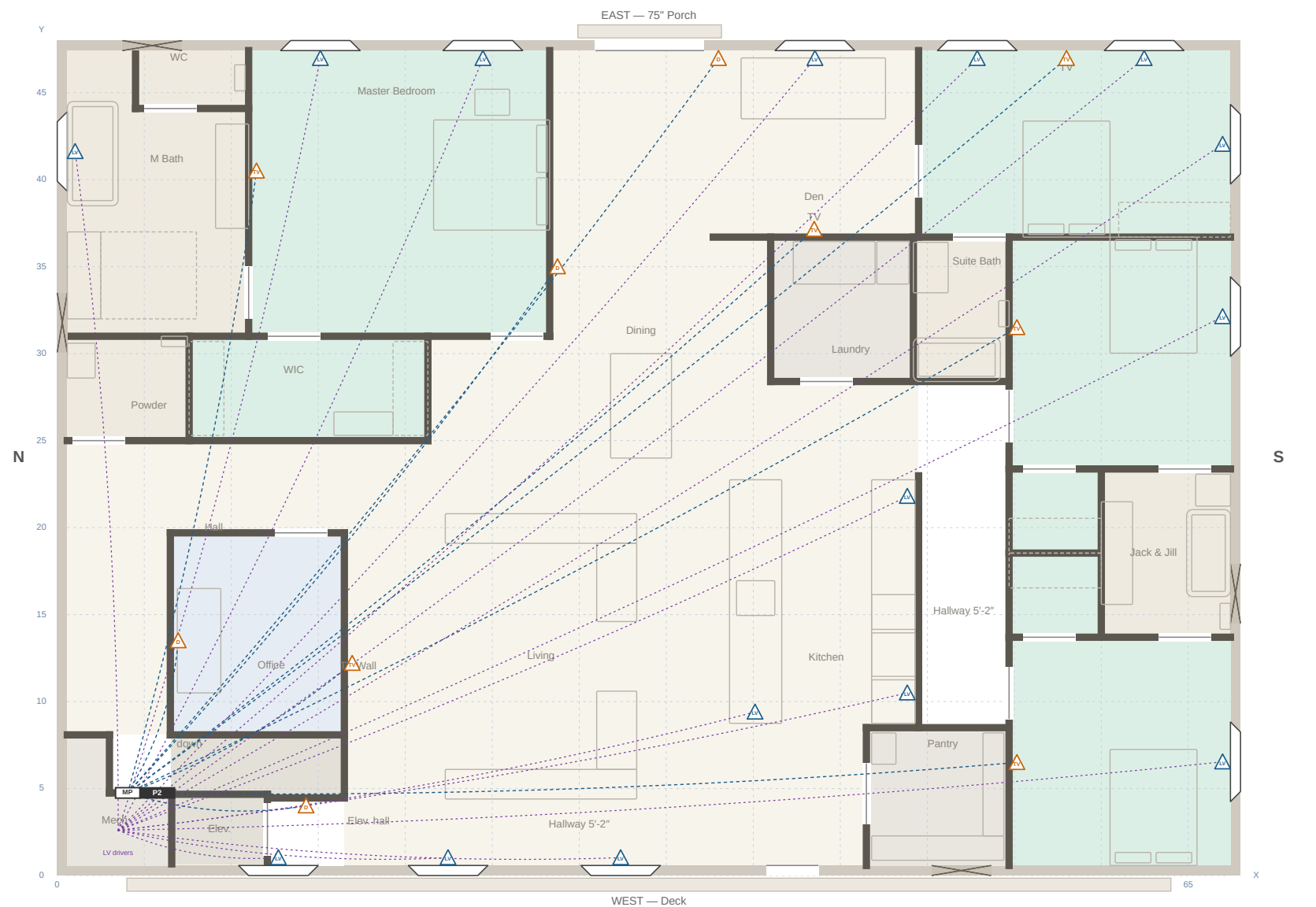
duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d	
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29	Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson
E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.26 — circuit 30: Low-voltage power supplies, 15A, #14 · shade and LED drivers in the mech room



duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d			
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29		Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson	
		E-1 rev C · 2026-10-01 drawn for K. Christianson	

E-1.D — Cat6, coax and low-voltage home runs



duplex receptacle 18" GFCI 20A 42" pop-up counter 240V TV box switch S / S3 / SD dimmer / So occupancy recessed LED pendant sconce exhaust fan smoke / smoke+CO soffit light low-voltage d	
7824 Zero Road — second floor Casper, Wyoming · 68' × 48' to brick, faces 7" inside · 9' ceiling · grid 5' Wall numbers as the floor plan drawing of 2026-09-29	Electrical plan Devices, luminaires, alarms and the circuit schedule for sub-panel P2 Schematic for pricing and coordination, not a stamped construction document. Verify on site for K. Christianson

E-1

rev C · 2026-10-01

drawn for K. Christianson

Notes

1. Code basis: NEC 2023 (confirm the edition adopted by Casper / Natrona County). AFCI on every 120V 15A and 20A branch circuit except the bathroom receptacle circuits (210.12); GFCI at bathrooms, kitchen counters, laundry, dishwasher, exterior and within 6' of any sink (210.8).
2. Accessible mounting: every wall receptacle, 240V outlet and data outlet 30" AFF to centre; switches, dimmers and the GFCI receptacles above counters and vanities 42" AFF (reach range 15"—48"). Rocker switches throughout; no device above a counter deeper than 20".
3. Switch plan: every light belongs to one switch leg (table below). On the page, clicking a light highlights every light and switch on its leg; clicking a switch highlights every leg it serves. A switch tag that appears in several legs is a multi-gang bank, one rocker per leg.
4. Circuit sheets E-1.1 onward show one circuit each with its cable run drawn as a chain from P2 through every device in nearest-neighbour order (solid for receptacle and appliance circuits, dashed for lighting). The run is a proposed order, not a required one. E-1.D shows the Cat6 home runs to the media panel beside P2.
5. Sub-panel P2 and the media panel face the landing on wall 1. Bedroom 3's TV moves to wall 7: wall 10 is the closet door and the Jack & Jill pocket from end to end.
6. Kitchen: two 20A small-appliance circuits (7, 8). Island receptacles are pop-up units in the counter at the sink end and the prep end plus one in each end panel; the wall run has counter receptacles at 42". Cooktop, oven, refrigerator/freezer, dishwasher drawers and microwave drawer each on a dedicated circuit. Hood ducted 6" up through the roof.
7. Lighting: 4" recessed LED cans, 2700K, dimmable, on 0–10V or Lutron Caséta-type dimmers; wet-rated cans over the shower pad and tubs; closet fixtures surface LED with 12" clearance to shelves (410.16). Pendants over the island and the dining table. Hallway, stair and the great room on 3-way or multi-location dimmers.
8. Smoke alarms in every bedroom and the office; smoke + CO combination alarms outside each sleeping area (hall at the office, outside the master suite, wing hall, den). All 120V, interconnected, battery backup (IRC R314, R315).
9. Bath exhaust fans 110 CFM with humidity sensing, ducted to the exterior in rigid duct: powder and master bath to the north wall, WC to the east wall, suite bath and Jack & Jill up through the roof. Dryer: 4" rigid metal, up through the roof, 35' equivalent max.
10. TV locations get a recessed box with power and 2 × Cat6 at 60" AFF (no coax: everything streams); data outlets home-run to the media panel beside P2 on the landing (Cat6 to every TV, the office desk and the elevator hall for the controller).
11. Exterior soffits: LED downlights the length of the west soffit on one switch inside the deck door; on the east, three lights over the exit door on switch leg A and the rest of the east soffit on leg B, both rockers inside the east door (circuit 26). Weatherproof GFCI receptacles in the soffit at both ends of the west wall, at the deck door, and beside the east door (circuit 24).
12. Low voltage (purple on E-1.D): an 18/2 drop in 1/2" conduit at the head of every unblinded window for motorised shades (pull 18/4 instead where the shade brand wants wired control), at the island base for LED and charging, and at both kitchen cabinet runs for under-cabinet LED, all home-run to a driver board in the mech room on circuit 30. Cat6 also to the east door (doorbell, camera, lock at 48") and to a 2-gang touchscreen box on the dining face of wall 19 at Y 35', 48" AFF.
13. Kitchen: the induction cooktop is in the island (circuit 1-2, junction in the island base) with a downdraft vent whose blower and 6" duct run in the floor to the west wall (P-1). The wall run keeps the oven and the refrigerator/freezer plus a tall cabinet.

14. Elevator: 240V 30A and 120V 15A (car light and hoistway) from the main panel with a lockable disconnect at the machine space, telephone line in the car, all by the elevator contractor. Not on P2.

Open items

- Confirm the feeder route from the main panel to the mech closet, and the main panel's spare capacity (the 125A sub-panel assumes about 20 kVA of new load).
- Lighting control: simple dimmers as drawn, or a smart-home system (Caséta, RadioRA 3) with keypads at the bedside and the entry?
- Ceiling fans: fan-rated boxes are shown in the master, 2nd Suite and Bedrooms 3 and 4. Keep, or drop in favour of the cans?
- Receptacle finish and colour, and whether the kitchen pop-ups should include USB (nightstands get USB-C).
- Low-voltage: any ceiling speakers, and a camera at the west door to match the east.

Schedules

Circuit schedule, sub-panel P2

P2: 125A main-lug sub-panel in the mech closet, fed from the main panel below with #1 Cu (or 2/0 Al) THHN in 1-1/4" conduit, 4-wire, isolated neutral, bonded ground. Size by the NEC 220 load calculation; 125A assumed. The elevator keeps its own 240V 30A + 120V 15A feeds and disconnect from the main panel, by the elevator contractor.

Ckt	Load	Breaker	Wire	Serves	Devices
1 - 2	Induction cooktop	240V 40A 2P	#8 Cu	kitchen wall run	1
3 - 4	Wall oven, side-opening	240V 30A 2P	#10 Cu	kitchen wall run	1
5 - 6	Dryer	240V 30A 2P	#10 Cu	laundry	1
7	Kitchen small appliance 1	20A GFCI/AFCI	#12	island sink end, wall run	3
8	Kitchen small appliance 2 + pantry	20A GFCI/AFCI	#12	island prep end, wall run, pantry	4
9	Refrigerator	20A AFCI	#12	kitchen	1
10	Spare	20A	#12	—	0
11	Dishwasher drawers	20A GFCI/AFCI	#12	island	1
12	Microwave drawer	20A AFCI	#12	island	1
13	Laundry: washer, sink, counter	20A GFCI/AFCI	#12	laundry	3
14	Bath receptacles west	20A GFCI	#12	master bath, WC, powder	3
15	Bath receptacles east	20A GFCI	#12	suite bath, Jack & Jill	3
16	Master suite receptacles	15A AFCI	#14	master bedroom, walk-in	9
17	Bedrooms 3 and 4 receptacles	15A AFCI	#14	bedrooms 3, 4, closets, wing hall	15

Ckt	Load	Breaker	Wire	Serves	Devices
18	2nd Suite receptacles	15A AFCI	#14	2nd Suite	7
19	Living and dining receptacles	15A AFCI	#14	living, hallway, dining, east entry	15
20	Den, office, passage receptacles	15A AFCI	#14	den, office, passage, elevator hall	19
21	Lighting A	15A AFCI	#14	living, dining, kitchen, den, office, baths	58
22	Lighting B + alarms	15A AFCI	#14	bedrooms, closets, halls, stair, smoke/CO alarms	62
23	Exhaust fans + hood	15A AFCI	#14	five bath fans, range hood	6
24	Exterior receptacles	20A GFCI	#12	soffit receptacles, west and east	4
25	Mech closet: ventilation unit	15A	#14	ERV / vents, service receptacle	3
26	Exterior soffit lighting	15A	#14	west soffit, east door group, east soffit; three switch legs	21
27 - 28	Spare 2P	240V 30A 2P	—	future (sauna, EV on deck, etc.)	0
29	Spare	20A	#12	—	0
30	Low-voltage power supplies	15A	#14	shade and LED drivers in the mech room	15

Switch plan

One row per switch leg. Click a row to light up its switch and lights on the drawing.

Leg	What	Room	Switch(es)	Lights	Control	Notes
stair	Stair and landing	stair	S1 3-way switch (stair light, top; 3-way with the bottom of the stair (floor below))	L4 4" recessed LED, L5 4" recessed LED, L3 4" recessed LED	3-way with a switch at the bottom of the stair (floor below)	the elevator-hall can rides this leg so the arrival is never dark
mech	Mech room	mech room	—	L1 surface LED ceiling fixture, L2 surface LED ceiling fixture	integral occupancy sensors, no switch	the room has no door
office	Office cans	office	S2 dimmer (office door)	L6 4" recessed LED, L7 4" recessed LED, L8 4" recessed LED, L9 4" recessed LED	dimmer at the office door	
passage	Passage and hall cans	passage, hall	S3 3-way switch (passage lights, stair end); S4 3-way switch (passage lights, dining end)	L10 4" recessed LED, L11 4" recessed LED, L12 4" recessed LED, L13 4" recessed LED, L14 4" recessed LED	3-way: stair end and dining end	
powder - light	Powder room light and sconce	powder	S5 switch (powder: light + fan, two rockers)	L15 4" recessed LED, L16 wall sconce / vanity light	rocker 1 of a 2-gang at the door	
powder - fan	Powder room fan	powder	S5 switch (powder: light + fan, two rockers)	L17 exhaust fan	rocker 2, with a 20-minute timer	

Leg	What	Room	Switch(es)	Lights	Control	Notes
walkin	Walk-in closet	walk-in	S6 occupancy-sensor switch (walk-in, beyond the pocket of D12)	L18 surface LED, L19 surface LED	occupancy-sensor switch	
master-cans	Master bedroom cans	master bedroom	S7 dimmer (at the bedroom pocket door D13, strike side; 3-way with the bedside); S8 3-way switch (bedside, reachable from the bed)	L20 4" recessed LED, L21 4" recessed LED, L22 4" recessed LED, L23 4" recessed LED	3-way dimmer: door and bedside	the bedside switch is reachable from the bed
master-fan	Master bedroom fan box	master bedroom	S7 dimmer (at the bedroom pocket door D13, strike side; 3-way with the bedside)	L24 fan-rated ceiling box	rocker 2 at the door; fan speed from the fan remote	
mbath-cans	Master bath general can	master bath	S9 dimmer (bath bank at the 6' opening: cans, shower, tub, sconces, fan (5 rockers))	L27 4" recessed LED	rocker 1 of the 5-gang bank at the bath door	
mbath-shower	Shower light	master bath	S9 dimmer (bath bank at the 6' opening: cans, shower, tub, sconces, fan (5 rockers))	L25 4" recessed LED	rocker 2	wet-location can over the pad
mbath-tub	Tub light	master bath	S9 dimmer (bath bank at the 6' opening: cans, shower, tub, sconces, fan (5 rockers))	L26 4" recessed LED	rocker 3, dimmer	
mbath-sconces	Vanity sconces	master bath	S9 dimmer (bath bank at the 6' opening: cans, shower, tub, sconces, fan (5 rockers))	L28 wall sconce / vanity light, L29 wall sconce / vanity light	rocker 4, dimmer	
mbath-fan	Master bath fan	master bath	S9 dimmer (bath bank at the 6' opening: cans, shower, tub, sconces, fan (5 rockers))	L30 exhaust fan	rocker 5, humidity sensing runs it anyway	
wc-light	WC light	WC	S10 switch (WC: light + fan)	L31 4" recessed LED	rocker 1 of a 2-gang inside the WC door	
wc-fan	WC fan	WC	S10 switch (WC: light + fan)	L32 exhaust fan	rocker 2, timer	
living	Living cans	living	S11 dimmer (living cans, elevator end; 3-way); S12 dimmer (living cans, deck door; 3-way; exterior light)	L33 4" recessed LED, L34 4" recessed LED, L35 4" recessed LED, L36 4" recessed LED, L37 4" recessed LED, L38 4" recessed LED	3-way dimmer: elevator end and deck door	
hall-west	West hallway cans	hallway	S11 dimmer (living cans, elevator end; 3-way); S12 dimmer (living cans, deck door; 3-way; exterior light)	L39 4" recessed LED, L40 4" recessed LED, L41 4" recessed LED, L42 4" recessed LED	3-way, rocker 2 at both ends	

Leg	What	Room	Switch(es)	Lights	Control	Notes
west-soffit	West soffit lights	exterior, deck side	S12 dimmer (living cans, deck door; 3-way; exterior light)	L91 exterior soffit downlight, L92 exterior soffit downlight, L93 exterior soffit downlight, L94 exterior soffit downlight, L95 exterior soffit downlight, L96 exterior soffit downlight, L97 exterior soffit downlight, L98 exterior soffit downlight, L99 exterior soffit downlight	rocker 3 inside the deck door; photocell or timer optional	circuit 26
dining-pendant	Dining pendant	dining	S13 dimmer (dining pendant + cans, on wall 19 at the master bedroom door)	L43 pendant	dimmer, rocker 1 on wall 19	
dining-cans	Dining and east entry cans	dining	S13 dimmer (dining pendant + cans, on wall 19 at the master bedroom door)	L44 4" recessed LED, L45 4" recessed LED, L46 4" recessed LED	dimmer, rocker 2	
east-door	East door soffit lights (3)	exterior, east door	S14 switch (east door: two rockers, three door lights and the whole east soffit)	L100 exterior soffit downlight, L101 exterior soffit downlight, L102 exterior soffit downlight	rocker A inside the east door	
east-soffit	Whole east soffit	exterior, east	S14 switch (east door: two rockers, three door lights and the whole east soffit)	L103 exterior soffit downlight, L104 exterior soffit downlight, L105 exterior soffit downlight, L106 exterior soffit downlight, L107 exterior soffit downlight, L108 exterior soffit downlight, L109 exterior soffit downlight, L110 exterior soffit downlight	rocker B inside the east door	the three door lights stay on rocker A; rocker B adds the rest
kitchen-pendants	Island pendants	kitchen	S15 dimmer (kitchen: pendants, cans, shelf light; 3-way); S16 3-way switch (kitchen, dining end beside the laundry door; 3-way)	L49 pendant, L50 pendant, L51 pendant	3-way dimmer: hall end and dining end	
kitchen-cans	Kitchen aisle cans	kitchen	S15 dimmer (kitchen: pendants, cans, shelf light; 3-way); S16 3-way switch (kitchen, dining end beside the laundry door; 3-way)	L52 4" recessed LED, L53 4" recessed LED, L54 4" recessed LED	3-way dimmer, rocker 2	
kitchen-shelf	Under-shelf strip	kitchen	S15 dimmer (kitchen: pendants, cans, shelf light; 3-way)	L48 linear LED under shelf / cabinet	rocker 3 at the hall end	
downdraft	Cooktop downdraft	kitchen	—	L47 downdraft cooktop vent	controls on the unit	no wall switch
pantry	Pantry light	pantry	S17 occupancy-sensor switch (pantry, inside the door)	L55 surface LED ceiling fixture	occupancy-sensor switch inside the opening	
wing-hall	Wing hall cans	wing hallway	S18 3-way switch (wing hall, kitchen end); S19 3-way switch (wing hall, dining end)	L56 4" recessed LED, L57 4" recessed LED, L58 4" recessed LED	3-way: kitchen end and dining end	
laundry	Laundry lights	laundry	S20 switch (laundry, inside the door)	L59 surface LED ceiling fixture, L60 surface LED ceiling fixture	single switch inside the door	

Leg	What	Room	Switch(es)	Lights	Control	Notes
suite-bath-light	Suite bath can and sconce	suite bath	S21 switch (suite bath: can, sconce, fan)	L61 4" recessed LED, L62 wall sconce / vanity light	rocker 1 of a 2-gang inside the door	
suite-bath-fan	Suite bath fan	suite bath	S21 switch (suite bath: can, sconce, fan)	L63 exhaust fan	rocker 2, timer	
den	Den cans	den	S22 dimmer (den cans)	L64 4" recessed LED, L65 4" recessed LED, L66 4" recessed LED, L67 4" recessed LED	dimmer at the den opening	
mini-cans	2nd Suite cans	2nd Suite	S23 dimmer (2nd Suite door; 3-way); S24 3-way switch (bedside)	L68 4" recessed LED, L69 4" recessed LED, L70 4" recessed LED, L71 4" recessed LED	3-way dimmer: door and bedside	
mini-closet	2nd Suite wardrobe light	2nd Suite	S23 dimmer (2nd Suite door; 3-way)	L72 surface LED	rocker 2 at the door	
mini-fan	2nd Suite fan box	2nd Suite	S24 3-way switch (bedside)	L73 fan-rated ceiling box	rocker 2 at the bedside; speed from the fan remote	
bed4-cans	Bedroom 4 cans	Bedroom 4	S25 dimmer (Bedroom 4 door)	L74 4" recessed LED, L75 4" recessed LED, L76 4" recessed LED, L77 4" recessed LED	dimmer at the door	
bed4-fan	Bedroom 4 fan box	Bedroom 4	S25 dimmer (Bedroom 4 door)	L78 fan-rated ceiling box	rocker 2 at the door	
closet4	Closet 4	closet 4	S26 occupancy-sensor switch (closet 4)	L79 surface LED	occupancy-sensor switch	
jj-lights	Jack & Jill lights	Jack & Jill	S27 3-way switch (Bedroom 3 door: lights + fan, 3-way); S28 3-way switch (Bedroom 4 door: lights + fan, 3-way)	L80 4" recessed LED, L81 4" recessed LED, L82 wall sconce / vanity light, L83 wall sconce / vanity light	3-way at both doors, rocker 1	
jj-fan	Jack & Jill fan	Jack & Jill	S27 3-way switch (Bedroom 3 door: lights + fan, 3-way); S28 3-way switch (Bedroom 4 door: lights + fan, 3-way)	L84 exhaust fan	3-way at both doors, rocker 2, timer	
bed3-cans	Bedroom 3 cans	Bedroom 3	S29 dimmer (Bedroom 3 door)	L85 4" recessed LED, L86 4" recessed LED, L87 4" recessed LED, L88 4" recessed LED	dimmer at the door	
bed3-fan	Bedroom 3 fan box	Bedroom 3	S29 dimmer (Bedroom 3 door)	L89 fan-rated ceiling box	rocker 2 at the door	
closet3	Closet 3	closet 3	S30 occupancy-sensor switch (closet 3)	L90 surface LED	occupancy-sensor switch	

Device schedule

Tag	Device	Position	AFF	Circuit	Notes
X1	sub-panel P2, 125A, 30 space	X 5'-4", Y 4'-9"	60"	—	Sub-panel P2 on wall 1, landing face, top of cabinet 72" AFF; 36" clear in front on the landing
X2	media panel: Cat6 home runs, network switch	X 3'-7", Y 4'-9"	54"	—	Media panel beside P2 on the landing face of wall 1: every Cat6 home-runs here
R1	receptacle, 20A dedicated	X 7", Y 2'-7"	30"	25	ventilation unit (ERV) service receptacle
L1	surface LED ceiling fixture	X 3'-6", Y 2'-7"	108"	25	mech room, fixture with integral occupancy sensor (no door, no switch)
L2	surface LED ceiling fixture	X 1'-10", Y 6'-5"	108"	25	vent leg, fixture with integral occupancy sensor
L3	4" recessed LED, 2700K, IC, dimmable	X 14'-4", Y 2'-5"	108"	22	elevator hall
R2	duplex receptacle, 15A	X 14'-4", Y 7"	30"	20	elevator hall
L4	4" recessed LED, 2700K, IC, dimmable	X 11'-6", Y 6'-5"	108"	22	over the stair
L5	4" recessed LED, 2700K, IC, dimmable	X 4'-9", Y 6'-5"	108"	22	landing
S1	3-way switch, rocker	X 3'-0", Y 5'-5"	42"	22	stair light, top; 3-way with the bottom of the stair (floor below)
L6	4" recessed LED, 2700K, IC, dimmable	X 9'-0", Y 11'-0"	108"	21	
L7	4" recessed LED, 2700K, IC, dimmable	X 9'-0", Y 16'-6"	108"	21	
L8	4" recessed LED, 2700K, IC, dimmable	X 14'-0", Y 11'-0"	108"	21	
L9	4" recessed LED, 2700K, IC, dimmable	X 14'-0", Y 16'-6"	108"	21	
S2	dimmer, rocker	X 11'-11", Y 19'-8"	42"	21	office door
R3	duplex receptacle, 15A	X 6'-6", Y 10'-6"	30"	20	
R4	data outlet, 2 × Cat6	X 6'-6", Y 13'-6"	30"	—	desk
R5	duplex receptacle, 15A	X 6'-6", Y 15'-6"	30"	20	
R6	duplex receptacle, 15A	X 16'-6", Y 10'-0"	30"	20	

Tag	Device	Position	AFF	Circuit	Notes
R7	duplex receptacle, 15A	X 16'-6", Y 15'-0"	30"	20	
R8	duplex receptacle, 15A	X 8'-6", Y 8'-1"	30"	20	
R9	duplex receptacle, 15A	X 13'-0", Y 8'-1"	30"	20	
R10	duplex receptacle, 15A	X 8'-6", Y 19'-8"	30"	20	
A1	smoke alarm, 120V interconnected w/ battery	X 11'-6", Y 13'-11"	108"	22	office (could serve as a bedroom)
L10	4" recessed LED, 2700K, IC, dimmable	X 3'-6", Y 11'-6"	108"	22	
L11	4" recessed LED, 2700K, IC, dimmable	X 3'-6", Y 17'-6"	108"	22	
L12	4" recessed LED, 2700K, IC, dimmable	X 6'-0", Y 22'-5"	108"	22	
L13	4" recessed LED, 2700K, IC, dimmable	X 12'-0", Y 22'-5"	108"	22	
L14	4" recessed LED, 2700K, IC, dimmable	X 18'-0", Y 22'-5"	108"	22	
S3	3-way switch, rocker	X 6'-6", Y 9'-0"	42"	22	passage lights, stair end
S4	3-way switch, rocker	X 21'-0", Y 25'-0"	42"	22	passage lights, dining end
R11	duplex receptacle, 15A	X 7", Y 12'-0"	30"	20	
R12	duplex receptacle, 15A	X 7", Y 21'-0"	30"	20	
R13	duplex receptacle, 15A	X 9'-0", Y 25'-0"	30"	20	
R14	duplex receptacle, 15A	X 18'-0", Y 25'-0"	30"	20	
A2	smoke + CO alarm, 120V interconnected w/ battery	X 9'-0", Y 22'-5"	108"	22	hall outside the office
L15	4" recessed LED, 2700K, IC, dimmable	X 4'-0", Y 27'-6"	108"	21	
L16	wall sconce / vanity light	X 7", Y 29'-7"	72"	21	over the powder lav
L17	exhaust fan, 110 CFM, humidity sensing	X 5'-6", Y 27'-6"	108"	23	powder, duct to the north wall

Tag	Device	Position	AFF	Circuit	Notes
S5	switch, rocker	X 4'-5", Y 25'-0"	42"	21	powder: light + fan, two rockers
R15	GFCI receptacle, 20A	X 7", Y 27'-10"	42"	14	42" AFF beside the lav
L18	surface LED, closet-rated (NEC 410.16)	X 11'-0", Y 28'-0"	108"	22	
L19	surface LED, closet-rated (NEC 410.16)	X 17'-0", Y 28'-0"	108"	22	
S6	occupancy-sensor switch	X 18'-8", Y 31'-0"	42"	22	walk-in, beyond the pocket of D12
R16	duplex receptacle, 15A	X 21'-4", Y 28'-0"	30"	16	
L20	4" recessed LED, 2700K, IC, dimmable	X 15'-0", Y 35'-0"	108"	22	
L21	4" recessed LED, 2700K, IC, dimmable	X 15'-0", Y 43'-0"	108"	22	
L22	4" recessed LED, 2700K, IC, dimmable	X 23'-0", Y 34'-6"	108"	22	
L23	4" recessed LED, 2700K, IC, dimmable	X 23'-0", Y 45'-6"	108"	22	
L24	fan-rated ceiling box, switched	X 19'-7", Y 39'-2"	108"	22	room centre
S7	dimmer, rocker	X 21'-2", Y 31'-0"	42"	22	at the bedroom pocket door D13, strike side; 3-way with the bedside
S8	3-way switch, rocker	X 28'-4", Y 36'-5"	42"	22	bedside, reachable from the bed
R17	duplex receptacle with USB-C, 15A	X 28'-4", Y 35'-0"	30"	16	nightstand
R18	duplex receptacle with USB-C, 15A	X 28'-4", Y 45'-6"	30"	16	nightstand
R19	duplex receptacle, 15A	X 12'-0", Y 47'-5"	30"	16	
R20	duplex receptacle, 15A	X 19'-0", Y 47'-5"	30"	16	
R21	duplex receptacle, 15A	X 27'-6", Y 47'-5"	30"	16	
R22	recessed TV box: power + 2 × Cat6	X 11'-0", Y 40'-6"	60"	16	TV on wall 23, 60" AFF

Tag	Device	Position	AFF	Circuit	Notes
R23	duplex receptacle, 15A	X 11'-0", Y 44'-6"	30"	16	
R24	duplex receptacle, 15A	X 20'-0", Y 31'-0"	30"	16	
A3	smoke + CO alarm, 120V interconnected w/ battery	X 16'-0", Y 39'-2"	108"	22	master bedroom
L25	4" recessed LED, wet location (shower / tub)	X 5'-2", Y 34'-6"	108"	21	over the shower pad
L26	4" recessed LED, wet location (shower / tub)	X 2'-0", Y 41'-6"	108"	21	over the tub
L27	4" recessed LED, 2700K, IC, dimmable	X 6'-6", Y 40'-6"	108"	21	
L28	wall sconce / vanity light	X 11'-0", Y 38'-7"	72"	21	vanity, left, beyond the D18 pocket
L29	wall sconce / vanity light	X 11'-0", Y 42'-2"	72"	21	vanity, right
L30	exhaust fan, 110 CFM, humidity sensing	X 5'-0", Y 38'-6"	108"	23	master bath, between shower and tub; duct to the north wall
S9	dimmer, rocker	X 10'-7", Y 31'-0"	42"	21	bath bank at the 6' opening: cans, shower, tub, sconces, fan (5 rockers)
R25	GFCI receptacle, 20A	X 11'-0", Y 40'-2"	42"	14	vanity, 42" AFF, within 3' of both basins
L31	4" recessed LED, 2700K, IC, dimmable	X 7'-8", Y 45'-10"	108"	21	WC
L32	exhaust fan, 110 CFM, humidity sensing	X 6'-2", Y 46'-4"	108"	23	WC, duct to the east wall
S10	switch, rocker	X 8'-7", Y 44'-1"	42"	21	WC: light + fan
R26	GFCI receptacle, 20A	X 9'-4", Y 47'-5"	42"	14	bidet seat, beside the toilet
L33	4" recessed LED, 2700K, IC, dimmable	X 21'-0", Y 8'-0"	108"	21	
L34	4" recessed LED, 2700K, IC, dimmable	X 21'-0", Y 15'-0"	108"	21	
L35	4" recessed LED, 2700K, IC, dimmable	X 21'-0", Y 22'-0"	108"	21	

Tag	Device	Position	AFF	Circuit	Notes
L36	4" recessed LED, 2700K, IC, dimmable	X 30'-0", Y 8'-0"	108"	21	
L37	4" recessed LED, 2700K, IC, dimmable	X 30'-0", Y 15'-0"	108"	21	
L38	4" recessed LED, 2700K, IC, dimmable	X 30'-0", Y 22'-0"	108"	21	
L39	4" recessed LED, 2700K, IC, dimmable	X 22'-0", Y 3'-0"	108"	21	
L40	4" recessed LED, 2700K, IC, dimmable	X 30'-0", Y 3'-0"	108"	21	
L41	4" recessed LED, 2700K, IC, dimmable	X 38'-0", Y 3'-0"	108"	21	
L42	4" recessed LED, 2700K, IC, dimmable	X 44'-6", Y 3'-0"	108"	21	
S11	dimmer, rocker	X 16'-6", Y 5'-4"	42"	21	living cans, elevator end; 3-way
S12	dimmer, rocker	X 44'-4", Y 7"	42"	21	living cans, deck door; 3-way; exterior light
R27	weatherproof GFCI receptacle, in-use cover	X 44'-6", Y -4"	24"	24	west soffit receptacle at the deck door
R28	duplex receptacle, 15A	X 19'-0", Y 7"	30"	19	
R29	duplex receptacle, 15A	X 27'-0", Y 7"	30"	19	
R30	duplex receptacle, 15A	X 36'-0", Y 7"	30"	19	
R31	duplex receptacle, 15A	X 45'-6", Y 7"	30"	19	
R32	duplex receptacle, 15A	X 16'-6", Y 10'-0"	30"	19	
R33	recessed TV box: power + 2 × Cat6	X 16'-6", Y 12'-2"	60"	19	living TV centred on wall 27
R34	duplex receptacle, 15A	X 16'-6", Y 16'-0"	30"	19	
R35	floor box receptacle	X 27'-0", Y 5'-0"	—	19	under the north sofa
R36	floor box receptacle	X 27'-0", Y 19'-0"	—	19	under the south sofa
A4	smoke + CO alarm, 120V interconnected w/ battery	X 22'-0", Y 27'-6"	108"	22	outside the master suite door
L43	pendant	X 33'-6", Y 27'-0"	108"	21	over the table, dimmed

Tag	Device	Position	AFF	Circuit	Notes
L44	4" recessed LED, 2700K, IC, dimmable	X 30'-0", Y 33'-6"	108"	21	
L45	4" recessed LED, 2700K, IC, dimmable	X 36'-6", Y 33'-6"	108"	21	
L46	4" recessed LED, 2700K, IC, dimmable	X 33'-6", Y 42'-0"	108"	21	east entry
S13	dimmer, rocker	X 28'-4", Y 31'-10"	42"	21	dining pendant + cans, on wall 19 at the master bedroom door
S14	switch, rocker	X 30'-2", Y 47'-5"	42"	26	east door: two rockers, three door lights and the whole east soffit
R37	weatherproof GFCI receptacle, in-use cover	X 38'-10", Y 48'-4"	24"	24	east soffit receptacle beside the door
R38	duplex receptacle, 15A	X 28'-4", Y 34'-0"	30"	19	
R39	duplex receptacle, 15A	X 28'-4", Y 41'-0"	30"	19	
R40	duplex receptacle, 15A	X 28'-4", Y 46'-0"	30"	19	
R41	duplex receptacle, 15A	X 29'-6", Y 47'-5"	30"	19	
R42	duplex receptacle, 15A	X 41'-0", Y 30'-0"	30"	19	
R43	duplex receptacle, 15A	X 41'-0", Y 35'-0"	30"	19	
R44	pop-up counter receptacle, 20A GFCI	X 40'-1", Y 10'-0"	34"	7	island, sink end
R45	pop-up counter receptacle, 20A GFCI	X 40'-1", Y 21'-6"	34"	8	island, prep end
R46	duplex receptacle, 15A	X 40'-1", Y 8'-7"	30"	7	island end panel
R47	duplex receptacle, 15A	X 40'-1", Y 22'-11"	30"	8	island end panel
R48	receptacle, 20A dedicated	X 40'-1", Y 16'-0"	30"	11	dishwasher drawers, under the island
R49	receptacle, 20A dedicated	X 40'-1", Y 18'-4"	30"	12	microwave drawer, island

Tag	Device	Position	AFF	Circuit	Notes
R50	receptacle, 20A dedicated	X 49'-6", Y 10'-0"	30"	9	refrigerator / freezer, recessed box
R51	240V receptacle or junction	X 49'-6", Y 15'-2"	30"	3-4	wall oven, side-opening
R52	240V receptacle or junction	X 40'-1", Y 20'-4"	30"	1-2	induction cooktop in the island, junction below the counter
R53	GFCI receptacle, 20A	X 49'-6", Y 16'-11"	42"	7	counter, 42" AFF
R54	GFCI receptacle, 20A	X 49'-6", Y 22'-2"	42"	8	counter, 42" AFF
L47	downdraft cooktop vent, blower in the island, ducted in the floor	X 40'-1", Y 21'-7"	—	23	downdraft blower in the island, 6" duct in the floor to the west wall (P-1)
L48	linear LED under shelf / cabinet	X 49'-4", Y 20'-0"	54"	21	under the shelf, Y 17' to 23'
L49	pendant	X 40'-1", Y 12'-0"	108"	21	
L50	pendant	X 40'-1", Y 15'-6"	108"	21	
L51	pendant	X 40'-1", Y 19'-6"	108"	21	
L52	4" recessed LED, 2700K, IC, dimmable	X 44'-6", Y 11'-0"	108"	21	
L53	4" recessed LED, 2700K, IC, dimmable	X 44'-6", Y 16'-0"	108"	21	
L54	4" recessed LED, 2700K, IC, dimmable	X 44'-6", Y 21'-0"	108"	21	
S15	dimmer, rocker	X 46'-6", Y 7'-5"	42"	21	kitchen: pendants, cans, shelf light; 3-way
S16	3-way switch, rocker	X 46'-4", Y 28'-5"	42"	21	kitchen, dining end beside the laundry door; 3-way
L55	surface LED ceiling fixture	X 50'-7", Y 4'-6"	108"	21	pantry
S17	occupancy-sensor switch	X 46'-6", Y 7'-0"	42"	21	pantry, inside the door
R55	duplex receptacle, 15A	X 54'-8", Y 4'-6"	30"	8	pantry

Tag	Device	Position	AFF	Circuit	Notes
L56	4" recessed LED, 2700K, IC, dimmable	X 52'-1", Y 11'-0"	108"	22	
L57	4" recessed LED, 2700K, IC, dimmable	X 52'-1", Y 17'-6"	108"	22	
L58	4" recessed LED, 2700K, IC, dimmable	X 52'-1", Y 24'-0"	108"	22	
S18	3-way switch, rocker	X 52'-0", Y 8'-6"	42"	22	wing hall, kitchen end
S19	3-way switch, rocker	X 52'-0", Y 28'-5"	42"	22	wing hall, dining end
R56	duplex receptacle, 15A	X 54'-8", Y 16'-0"	30"	17	
A5	smoke + CO alarm, 120V interconnected w/ battery	X 52'-1", Y 20'-6"	108"	22	wing hall outside Bedrooms 3 and 4
R57	240V receptacle or junction	X 44'-8", Y 36'-8"	30"	5-6	dryer; recessed dryer box in wall 17 for the vent and cord
R58	receptacle, 20A dedicated	X 43'-5", Y 36'-8"	30"	13	washer
R59	GFCI receptacle, 20A	X 48'-0", Y 36'-8"	42"	13	laundry sink, 42" AFF
R60	duplex receptacle, 15A	X 41'-0", Y 31'-6"	30"	13	folding counter
L59	surface LED ceiling fixture	X 43'-6", Y 32'-0"	108"	22	
L60	surface LED ceiling fixture	X 47'-6", Y 32'-0"	108"	22	
S20	switch, rocker	X 46'-2", Y 28'-5"	42"	22	laundry, inside the door
L61	4" recessed LED, 2700K, IC, dimmable	X 52'-0", Y 33'-6"	108"	21	
L62	wall scone / vanity light	X 49'-2", Y 35'-0"	72"	21	over the suite vanity
L63	exhaust fan, 110 CFM, humidity sensing	X 52'-4", Y 31'-6"	108"	23	suite bath, duct up through the roof
S21	switch, rocker	X 50'-11", Y 36'-8"	42"	21	suite bath: can, scone, fan

Tag	Device	Position	AFF	Circuit	Notes
R61	GFCI receptacle, 20A	X 49'-2", Y 33'-2"	42"	15	suite vanity, 42" AFF
L64	4" recessed LED, 2700K, IC, dimmable	X 40'-6", Y 39'-6"	108"	21	
L65	4" recessed LED, 2700K, IC, dimmable	X 46'-6", Y 39'-6"	108"	21	
L66	4" recessed LED, 2700K, IC, dimmable	X 40'-6", Y 45'-0"	108"	21	
L67	4" recessed LED, 2700K, IC, dimmable	X 46'-6", Y 45'-0"	108"	21	
S22	dimmer, rocker	X 38'-4", Y 36'-8"	42"	21	den cans
R62	duplex receptacle, 15A	X 40'-0", Y 36'-8"	30"	20	
R63	recessed TV box: power + 2 × Cat6	X 43'-6", Y 36'-8"	60"	20	den TV on wall 17
R64	duplex receptacle, 15A	X 47'-0", Y 36'-8"	30"	20	
R65	duplex receptacle, 15A	X 49'-6", Y 38'-0"	30"	20	
R66	duplex receptacle, 15A	X 49'-6", Y 45'-6"	30"	20	
R67	duplex receptacle, 15A	X 39'-6", Y 47'-5"	30"	20	
R68	duplex receptacle, 15A	X 47'-10", Y 47'-5"	30"	20	
A6	smoke + CO alarm, 120V interconnected w/ battery	X 43'-6", Y 42'-0"	108"	22	den, outside the 2nd Suite
L68	4" recessed LED, 2700K, IC, dimmable	X 53'-6", Y 40'-6"	108"	22	
L69	4" recessed LED, 2700K, IC, dimmable	X 53'-6", Y 45'-0"	108"	22	
L70	4" recessed LED, 2700K, IC, dimmable	X 62'-6", Y 41'-0"	108"	22	
L71	4" recessed LED, 2700K, IC, dimmable	X 62'-6", Y 45'-0"	108"	22	

Tag	Device	Position	AFF	Circuit	Notes
L72	surface LED, closet-rated (NEC 410.16)	X 64'-0", Y 37'-8"	108"	22	wardrobe strip
L73	fan-rated ceiling box, switched	X 58'-0", Y 42'-0"	108"	22	
S23	dimmer, rocker	X 49'-6", Y 42'-7"	42"	22	2nd Suite door; 3-way
S24	3-way switch, rocker	X 54'-5", Y 36'-8"	42"	22	bedside
R69	duplex receptacle with USB-C, 15A	X 53'-6", Y 36'-8"	30"	18	nightstand
R70	duplex receptacle with USB-C, 15A	X 62'-0", Y 36'-8"	30"	18	nightstand
R71	duplex receptacle, 15A	X 49'-6", Y 45'-6"	30"	18	
R72	duplex receptacle, 15A	X 56'-0", Y 47'-5"	30"	18	
R73	recessed TV box: power + 2 × Cat6	X 58'-0", Y 47'-5"	60"	18	2nd Suite TV on the east wall
R74	duplex receptacle, 15A	X 66'-0", Y 47'-5"	30"	18	
R75	duplex receptacle, 15A	X 67'-5", Y 45'-6"	30"	18	
A7	smoke alarm, 120V interconnected w/ battery	X 58'-0", Y 44'-10"	108"	22	2nd Suite
L74	4" recessed LED, 2700K, IC, dimmable	X 57'-6", Y 27'-0"	108"	22	
L75	4" recessed LED, 2700K, IC, dimmable	X 57'-6", Y 34'-0"	108"	22	
L76	4" recessed LED, 2700K, IC, dimmable	X 65'-0", Y 27'-0"	108"	22	
L77	4" recessed LED, 2700K, IC, dimmable	X 65'-0", Y 34'-0"	108"	22	
L78	fan-rated ceiling box, switched	X 61'-0", Y 30'-0"	108"	22	
S25	dimmer, rocker	X 54'-8", Y 28'-7"	42"	22	Bedroom 4 door

Tag	Device	Position	AFF	Circuit	Notes
R76	duplex receptacle with USB-C, 15A	X 59'-0", Y 36'-8"	30"	17	nightstand
R77	duplex receptacle with USB-C, 15A	X 66'-7", Y 36'-8"	30"	17	nightstand
R78	duplex receptacle, 15A	X 54'-8", Y 33'-0"	30"	17	
R79	recessed TV box: power + 2 × Cat6	X 54'-8", Y 31'-6"	60"	17	TV on wall 7
R80	duplex receptacle, 15A	X 60'-0", Y 23'-4"	30"	17	
R81	duplex receptacle, 15A	X 67'-5", Y 26'-6"	30"	17	
R82	duplex receptacle, 15A	X 67'-5", Y 35'-6"	30"	17	
A8	smoke alarm, 120V interconnected w/ battery	X 61'-0", Y 32'-6"	108"	22	Bedroom 4
L79	surface LED, closet-rated (NEC 410.16)	X 57'-4", Y 21'-0"	108"	22	closet 4
S26	occupancy-sensor switch	X 59'-6", Y 23'-0"	42"	22	closet 4
L80	4" recessed LED, 2700K, IC, dimmable	X 63'-6", Y 18'-6"	108"	21	
L81	4" recessed LED, wet location (shower / tub)	X 66'-2", Y 18'-6"	108"	21	over the tub
L82	wall sconce / vanity light	X 60'-0", Y 16'-4"	72"	21	vanity, left
L83	wall sconce / vanity light	X 60'-0", Y 20'-10"	72"	21	vanity, right
L84	exhaust fan, 110 CFM, humidity sensing	X 64'-0", Y 15'-0"	108"	23	Jack & Jill, duct up through the roof
S27	3-way switch, rocker	X 61'-0", Y 14'-1"	42"	21	Bedroom 3 door: lights + fan, 3-way
S28	3-way switch, rocker	X 61'-0", Y 22'-11"	42"	21	Bedroom 4 door: lights + fan, 3-way
R83	GFCI receptacle, 20A	X 60'-0", Y 18'-7"	42"	15	between the basins, 42" AFF

Tag	Device	Position	AFF	Circuit	Notes
R84	GFCI receptacle, 20A	X 66'-0", Y 13'-8"	42"	15	bidet seat, beside the toilet
L85	4" recessed LED, 2700K, IC, dimmable	X 58'-0", Y 4'-0"	108"	22	
L86	4" recessed LED, 2700K, IC, dimmable	X 58'-0", Y 10'-0"	108"	22	
L87	4" recessed LED, 2700K, IC, dimmable	X 64'-6", Y 4'-0"	108"	22	
L88	4" recessed LED, 2700K, IC, dimmable	X 64'-6", Y 10'-0"	108"	22	
L89	fan-rated ceiling box, switched	X 61'-0", Y 7'-0"	108"	22	
S29	dimmer, rocker	X 54'-8", Y 12'-11"	42"	22	Bedroom 3 door
R85	duplex receptacle with USB-C, 15A	X 59'-0", Y 7"	30"	17	nightstand
R86	duplex receptacle with USB-C, 15A	X 67'-0", Y 7"	30"	17	nightstand
R87	duplex receptacle, 15A	X 54'-8", Y 3'-0"	30"	17	
R88	duplex receptacle, 15A	X 59'-4", Y 13'-8"	30"	17	
R89	recessed TV box: power + 2 × Cat6	X 54'-8", Y 6'-6"	60"	17	TV on wall 7 (wall 10 is all door and pocket now)
R90	duplex receptacle, 15A	X 67'-5", Y 2'-6"	30"	17	
R91	duplex receptacle, 15A	X 67'-5", Y 11'-0"	30"	17	
A9	smoke alarm, 120V interconnected w/ battery	X 61'-0", Y 9'-0"	108"	22	Bedroom 3
L90	surface LED, closet-rated (NEC 410.16)	X 57'-4", Y 16'-0"	108"	22	closet 3
S30	occupancy-sensor switch	X 59'-6", Y 14'-1"	42"	22	closet 3
R92	data outlet, 2 × Cat6	X 14'-4", Y 4'-6"	30"	—	elevator controller / phone line, on wall 6
R93	data outlet, 2 × Cat6	X 38'-0", Y 47'-5"	48"	—	east door: doorbell / camera / lock

Tag	Device	Position	AFF	Circuit	Notes
R94	data outlet, 2 × Cat6	X 28'-4", Y 35'-0"	48"	—	touchscreen insert on wall 19 (dining face), 2-gang box
X3	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 40'-1", Y 9'-5"	84"	30	island: LED and charging, up through the island base
X4	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 49'-4", Y 10'-6"	84"	30	kitchen cabinets, north run: under-cabinet LED driver lead
X5	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 49'-4", Y 21'-10"	84"	30	kitchen cabinets, south run: under-cabinet LED driver lead
X6	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 12'-9", Y 7"	90"	30	window head: motorised shade
X7	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 22'-6", Y 7"	90"	30	window head: motorised shade
X8	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 32'-5", Y 7"	90"	30	window head: motorised shade
X9	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 62'-6", Y 47'-5"	90"	30	window head: motorised shade
X10	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 52'-10", Y 47'-5"	90"	30	window head: motorised shade
X11	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 43'-6", Y 47'-5"	90"	30	window head: motorised shade
X12	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 24'-6", Y 47'-5"	90"	30	window head: motorised shade
X13	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 15'-1", Y 47'-5"	90"	30	window head: motorised shade
X14	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 7", Y 41'-7"	90"	30	window head: motorised shade
X15	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 67'-5", Y 6'-6"	90"	30	window head: motorised shade
X16	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 67'-5", Y 32'-2"	90"	30	window head: motorised shade
X17	low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	X 67'-5", Y 42'-0"	90"	30	window head: motorised shade
R95	weatherproof GFCI receptacle, in-use cover	X 1'-6", Y -4"	24"	24	west soffit receptacle, north end
R96	weatherproof GFCI receptacle, in-use cover	X 66'-6", Y -4"	24"	24	west soffit receptacle, south end
L91	exterior soffit downlight, LED, wet location	X 4'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L92	exterior soffit downlight, LED, wet location	X 12'-0", Y -4"	108"	26	west soffit, switched inside the deck door

Tag	Device	Position	AFF	Circuit	Notes
L93	exterior soffit downlight, LED, wet location	X 20'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L94	exterior soffit downlight, LED, wet location	X 28'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L95	exterior soffit downlight, LED, wet location	X 36'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L96	exterior soffit downlight, LED, wet location	X 46'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L97	exterior soffit downlight, LED, wet location	X 54'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L98	exterior soffit downlight, LED, wet location	X 60'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L99	exterior soffit downlight, LED, wet location	X 66'-0", Y -4"	108"	26	west soffit, switched inside the deck door
L100	exterior soffit downlight, LED, wet location	X 31'-6", Y 48'-4"	108"	26	east soffit over the door, switch leg A
L101	exterior soffit downlight, LED, wet location	X 34'-0", Y 48'-4"	108"	26	east soffit over the door, switch leg A
L102	exterior soffit downlight, LED, wet location	X 36'-6", Y 48'-4"	108"	26	east soffit over the door, switch leg A
L103	exterior soffit downlight, LED, wet location	X 4'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)
L104	exterior soffit downlight, LED, wet location	X 12'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)
L105	exterior soffit downlight, LED, wet location	X 20'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)
L106	exterior soffit downlight, LED, wet location	X 27'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)
L107	exterior soffit downlight, LED, wet location	X 41'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)
L108	exterior soffit downlight, LED, wet location	X 48'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)
L109	exterior soffit downlight, LED, wet location	X 56'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)
L110	exterior soffit downlight, LED, wet location	X 64'-0", Y 48'-4"	108"	26	east soffit, switch leg B (whole soffit)

Bill of materials

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The contractor takes off the final list.

Item	Qty	Notes
4" recessed LED, 2700K, IC, dimmable	56	
duplex receptacle, 15A	53	
exterior soffit downlight, LED, wet location	20	
low-voltage drop: 18/2 (and 18/4 control where the shade brand needs it) in 1/2" conduit to the driver board	15	
dimmer, rocker	11	
3-way switch, rocker	10	
GFCI receptacle, 20A	9	
duplex receptacle with USB-C, 15A	8	
wall sconce / vanity light	6	
recessed TV box: power + 2 × Cat6	6	
receptacle, 20A dedicated	5	
surface LED ceiling fixture	5	
smoke + CO alarm, 120V interconnected w/ battery	5	
exhaust fan, 110 CFM, humidity sensing	5	
switch, rocker	5	
surface LED, closet-rated (NEC 410.16)	5	
data outlet, 2 × Cat6	4	
smoke alarm, 120V interconnected w/ battery	4	
occupancy-sensor switch	4	
fan-rated ceiling box, switched	4	
weatherproof GFCI receptacle, in-use cover	4	
pendant	4	
4" recessed LED, wet location (shower / tub)	3	
240V receptacle or junction	3	
floor box receptacle	2	
pop-up counter receptacle, 20A GFCI	2	
sub-panel P2, 125A, 30 space	1	
media panel: Cat6 home runs, network switch	1	
downdraft cooktop vent, blower in the island, ducted in the floor	1	
linear LED under shelf / cabinet	1	

Item	Qty	Notes
Cable 8/3 NM-B	71 ft	chained run + 12' per device, +15%
Cable 10/3 NM-B	170 ft	chained run + 12' per device, +15%
Cable 12/2 NM-B	1090 ft	chained run + 12' per device, +15%
Cable 14/2 NM-B	5719 ft	chained run + 12' per device, +15%
14/3 NM-B for 3-way and multi-location switching	400 ft	40' per 3-way pair
Cat6 (two per TV/data outlet)	1532 ft	10 TV/data outlets
18/2 shade and LED cable in 1/2" ENT conduit	1329 ft	home runs to the mech room
Breakers 240V 40A 2P	1	AFCI/GFCI types as listed on the circuit schedule
Breakers 240V 30A 2P	3	AFCI/GFCI types as listed on the circuit schedule
Breakers 20A GFCI/AFCI	4	AFCI/GFCI types as listed on the circuit schedule
Breakers 20A AFCI	2	AFCI/GFCI types as listed on the circuit schedule
Breakers 20A	2	AFCI/GFCI types as listed on the circuit schedule
Breakers 20A GFCI	3	AFCI/GFCI types as listed on the circuit schedule
Breakers 15A AFCI	8	AFCI/GFCI types as listed on the circuit schedule
Breakers 15A	3	AFCI/GFCI types as listed on the circuit schedule
Sub-panel 125A MLO, 30 space, with #1 Cu feeder from the main	1	feeder length to confirm
Device boxes, old-work and new-work, with covers	262	4" square for 240V and TV boxes

Quantities are rough-order take-offs from the drawing geometry, for pricing. The contractor takes off the final list.